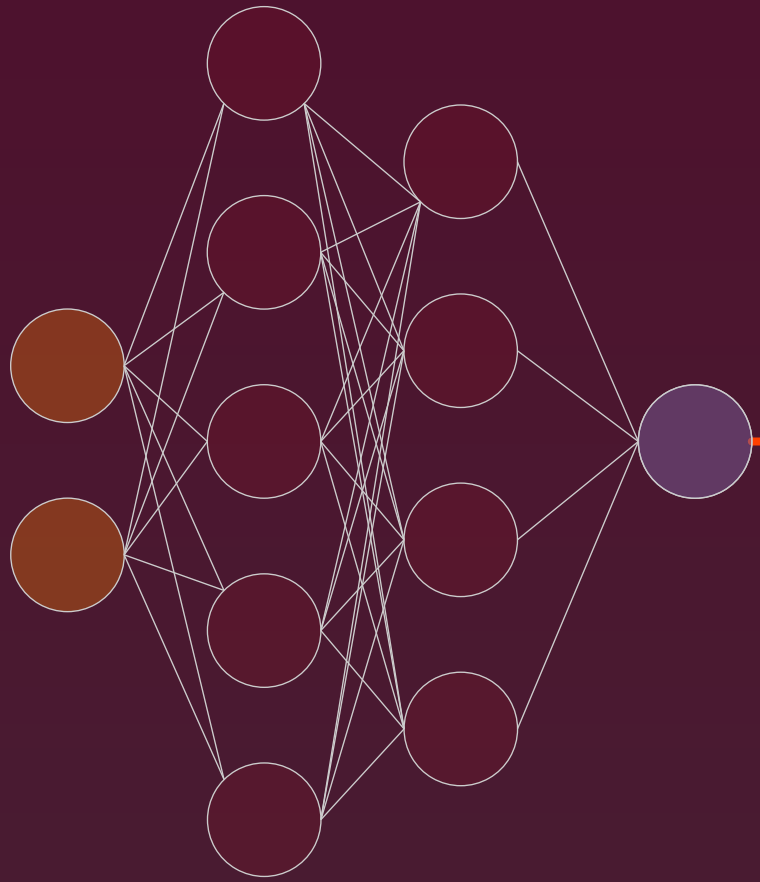
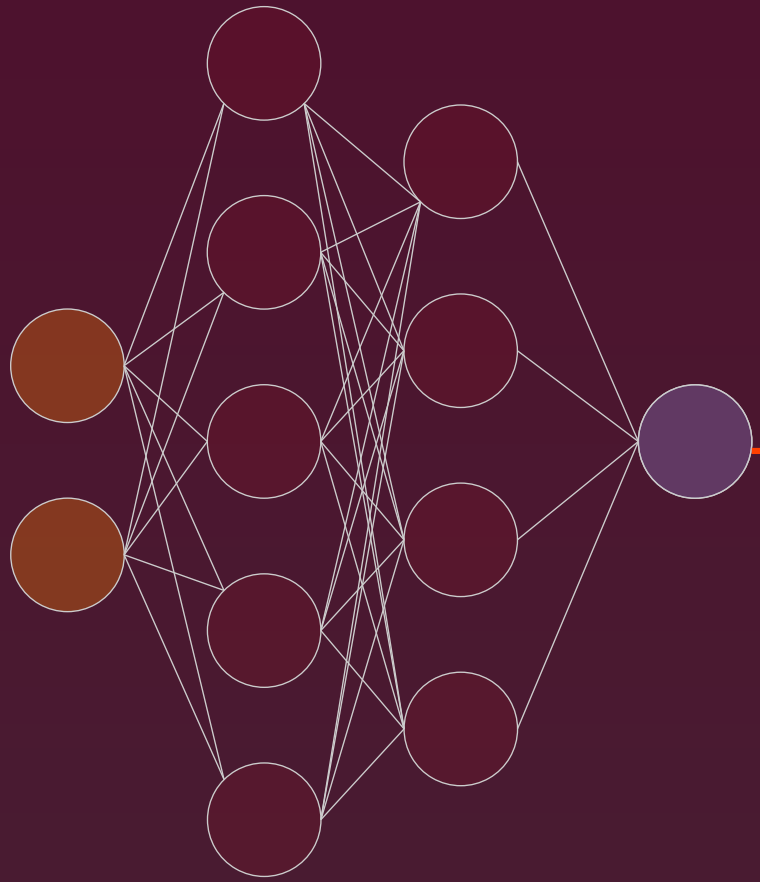


Réseaux de Neurones

By : Dr Nzamba Bignoumba



Programme

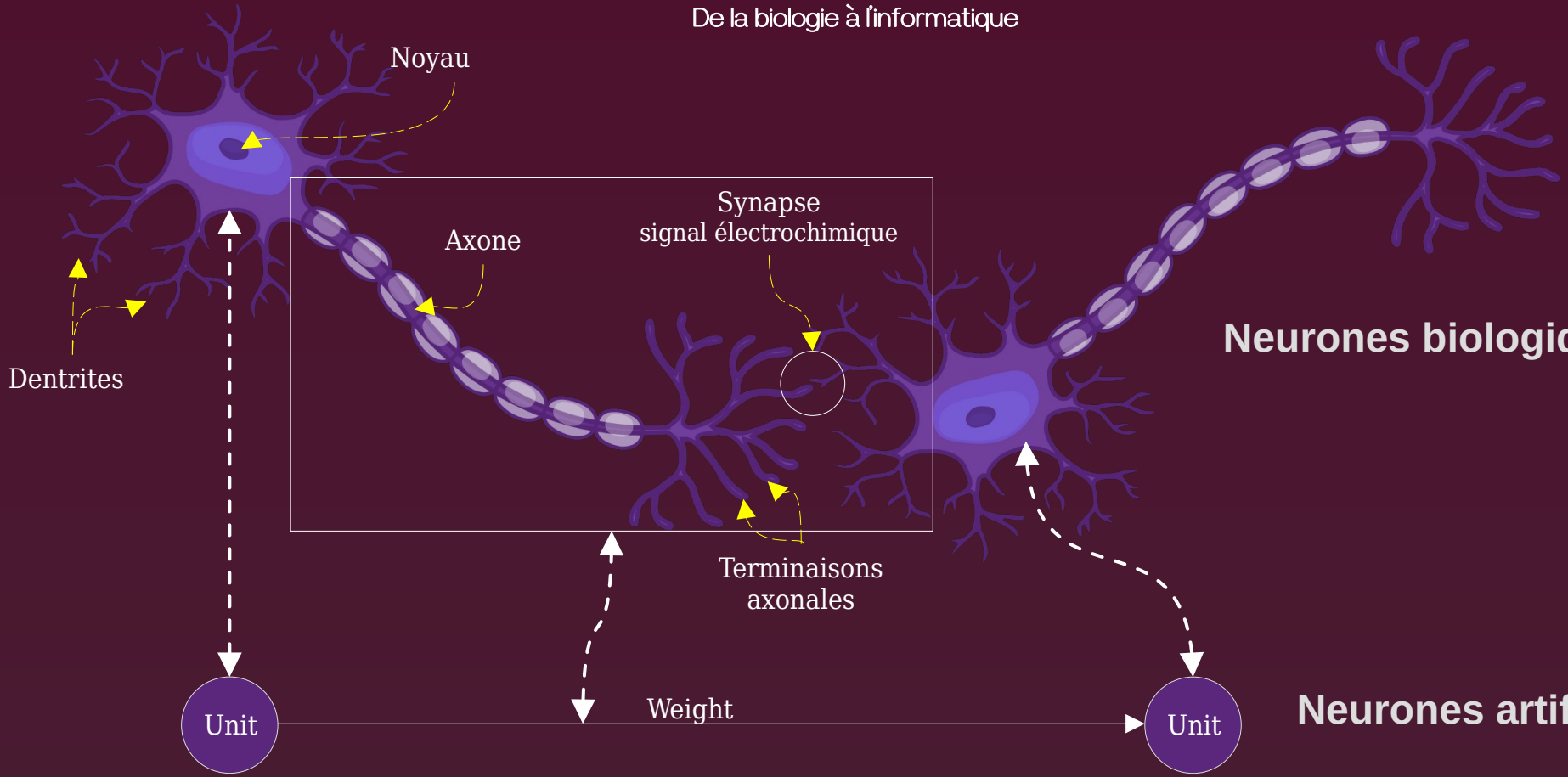
Introduction → 30 min

Théorie & Code → 6h

Déploiement du modèle → 1h30

Introduction

De la biologie à l'informatique



Neurones biologiques

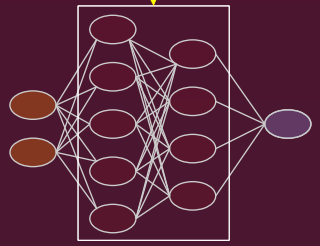
Neurones artificiels

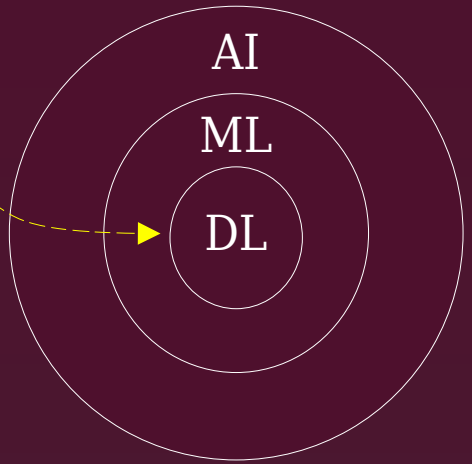
Introduction

Réseaux de neurones & deep learning

Réseaux de neurones : modèle informatique inspiré du fonctionnement du cerveau.

Adopte une méthode d'apprentissage fondée sur l'utilisation de couches superposées appelées hidden layers. Cette méthode d'apprentissage est appelée deep learning (apprentissage profond).





Réseaux de neurones
Type de tâches

Classification

- Détection de cancer
- Classification de plantes malades
- Reconnaissance d'images
- Détection de spam
- Analyse de sentiments
- Etc

Régression

- Prédiction du prix d'une maison
- Prévision des ventes
- Prévision météorologique
- Estimation de la consommation d'énergie
- Prédiction du taux d'occupation d'un parking
- Etc

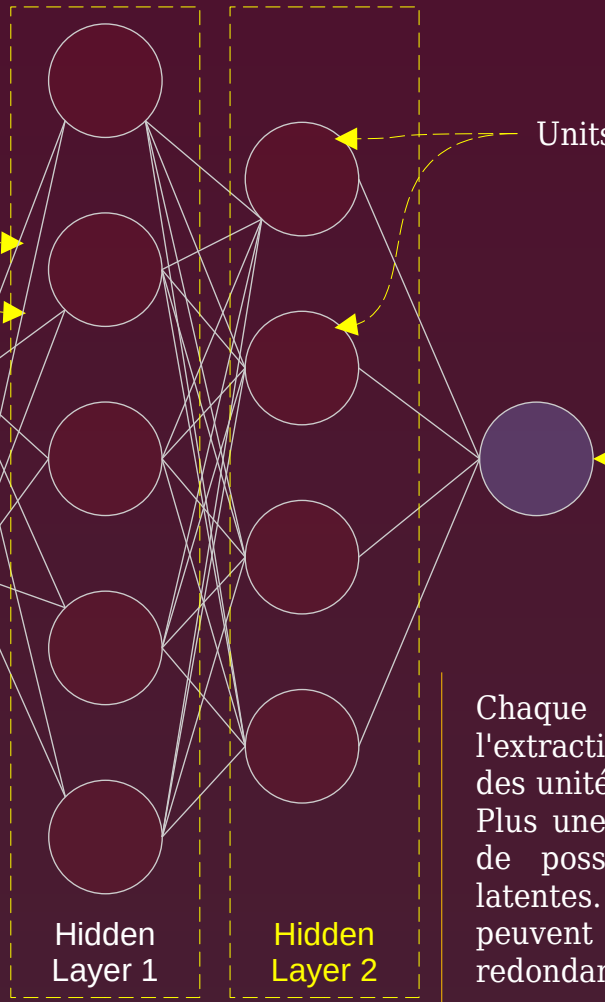
Structure d'un réseau de neurone

Paramètres du modèle (valeurs numériques). Ils seront ajustés au cours de la phase d'entraînement.

Weights

Les couches d'entrée contiennent des données brutes. Par exemple, il pourrait s'agir d'un vecteur $x = [\text{surface}, \text{matériau}]^T$ lorsque la tâche en aval consiste à prédire le loyer.

Input



Units

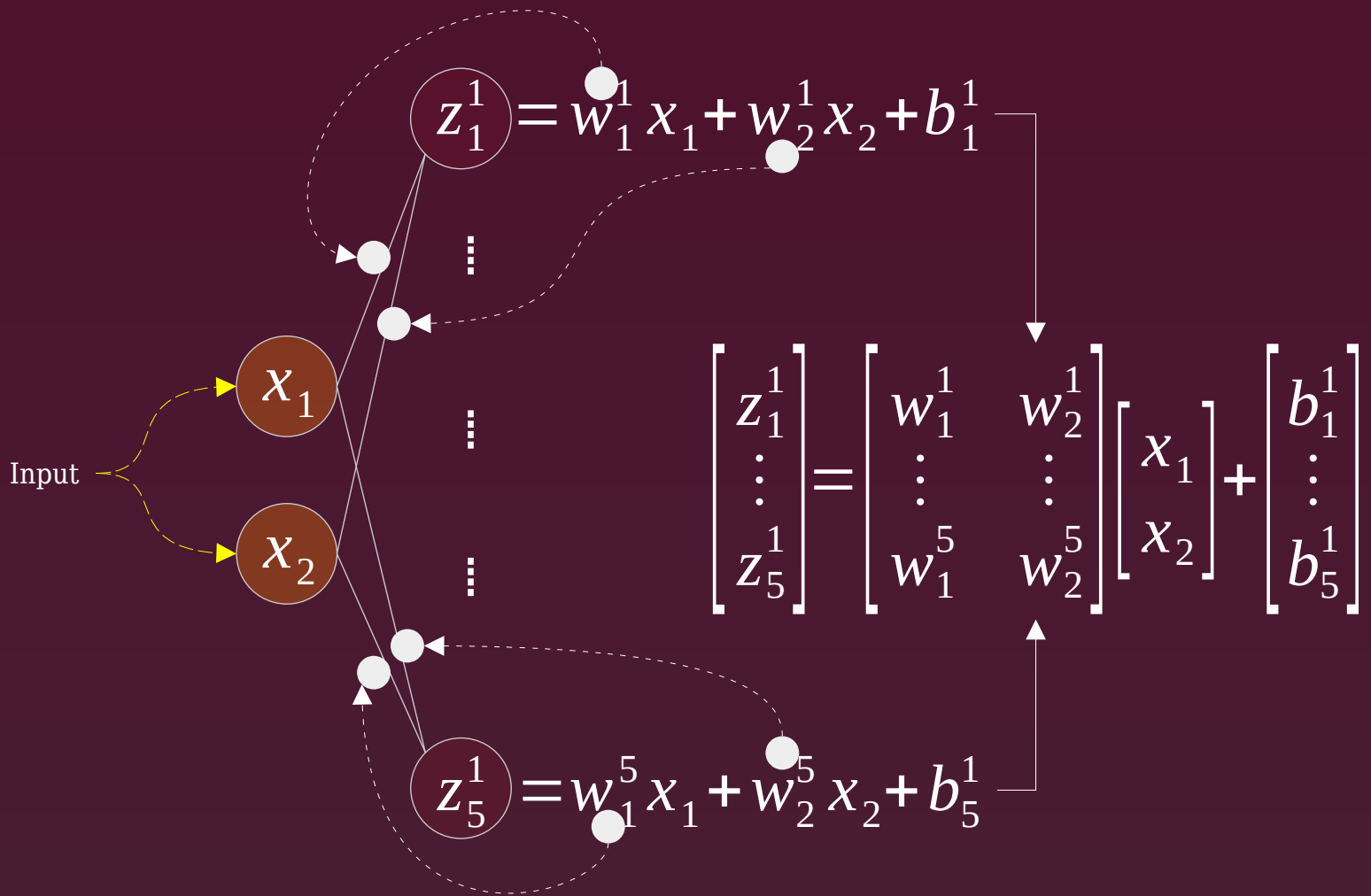
Unités de calculs. Extraits des caractéristiques latentes des unités de la couche précédente.

Output

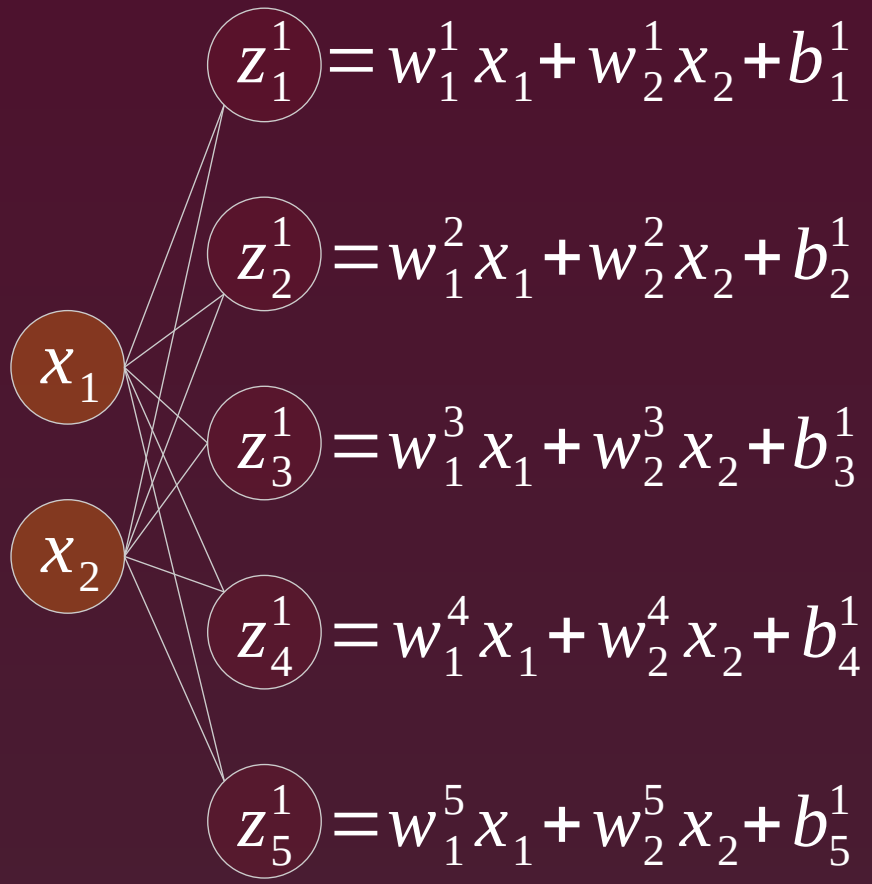
Couche de sortie. Comme nous avons pris l'exemple de la prédiction de loyer, la sortie est un nombre réel positif. Dans le jargon, ce type de tâche est appelé une tâche de régression

Chaque unité (Unit) est responsable de l'extraction des caractéristiques latentes issues des unités de la couche (hidden layer) précédente. Plus une couche compte d'unités, plus vous avez de possibilités d'extraire des caractéristiques latentes. Cependant, certaines d'entre elles peuvent constituer des caractéristiques latentes redondantes.

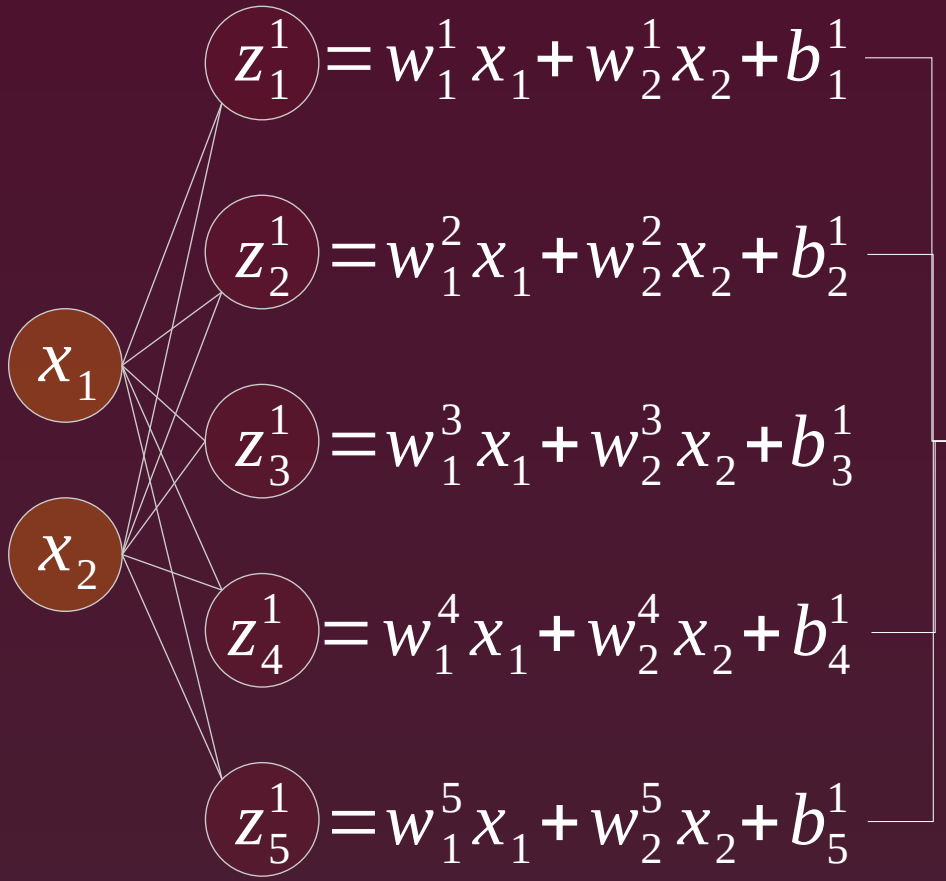
Forward pass



Forward pass



Forward pass

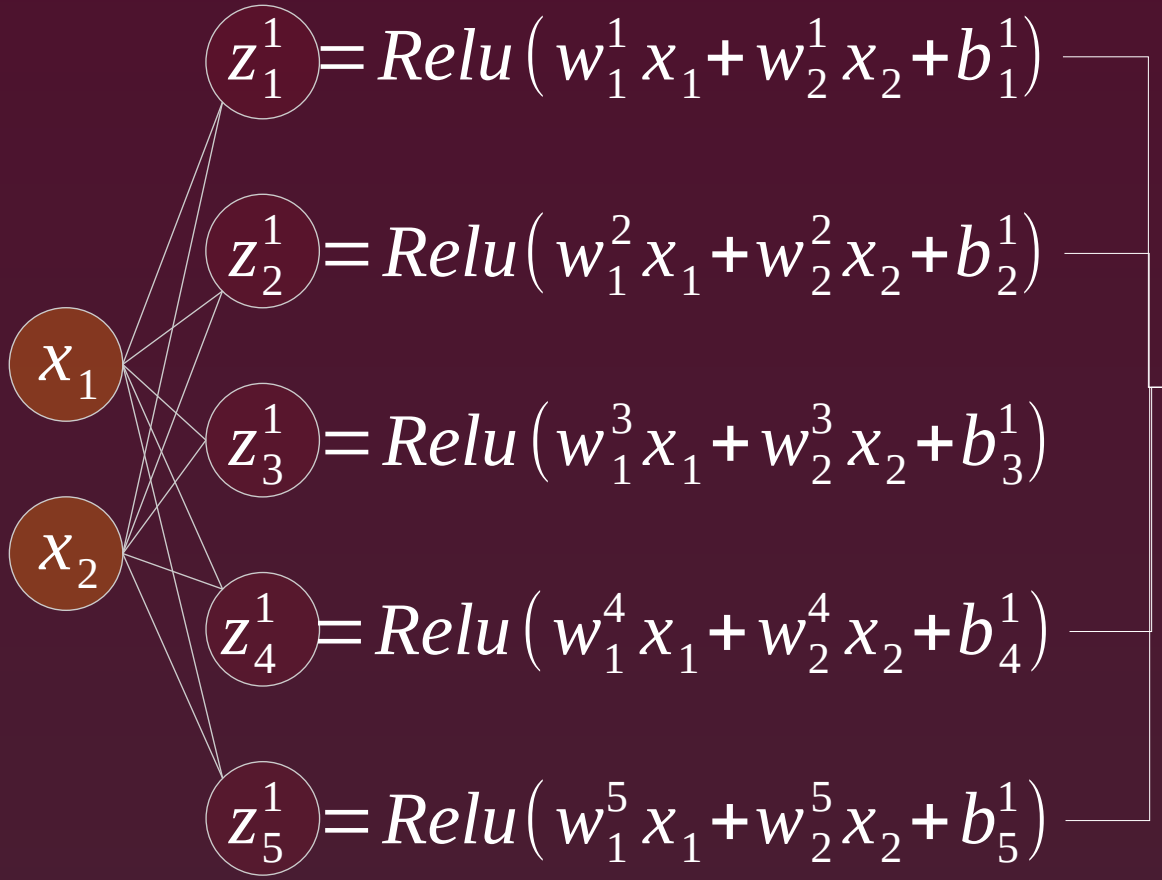


$$\begin{bmatrix} z_1^1 \\ z_2^1 \\ z_3^1 \\ z_4^1 \\ z_5^1 \end{bmatrix} = \begin{bmatrix} w_1^1 & w_2^1 \\ w_1^2 & w_2^2 \\ w_1^3 & w_2^3 \\ w_1^4 & w_2^4 \\ w_1^5 & w_2^5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1^1 \\ b_2^1 \\ b_3^1 \\ b_4^1 \\ b_5^1 \end{bmatrix}$$

Forward pass

Forward pass

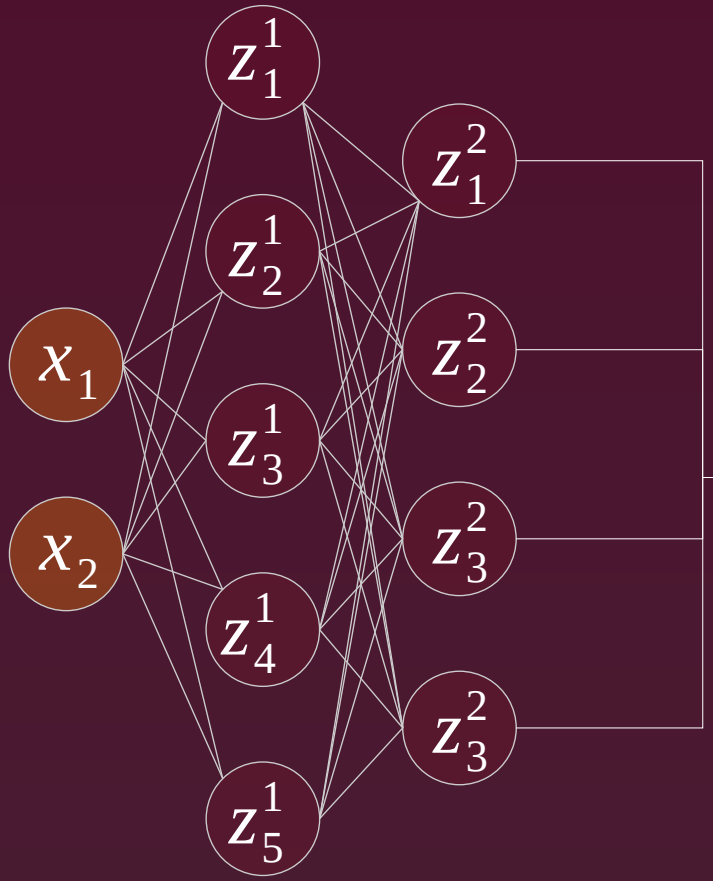
Utilisons une fonction non linéaire, à savoir ReLU().



$$\begin{bmatrix} Z_1^1 \\ Z_2^1 \\ Z_3^1 \\ Z_4^1 \\ Z_5^1 \end{bmatrix} = Relu\left(\begin{bmatrix} w_1^1 & w_2^1 \\ w_1^2 & w_2^2 \\ w_1^3 & w_2^3 \\ w_1^4 & w_2^4 \\ w_1^5 & w_2^5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1^1 \\ b_2^1 \\ b_3^1 \\ b_4^1 \\ b_5^1 \end{bmatrix} \right)$$

Forward pass

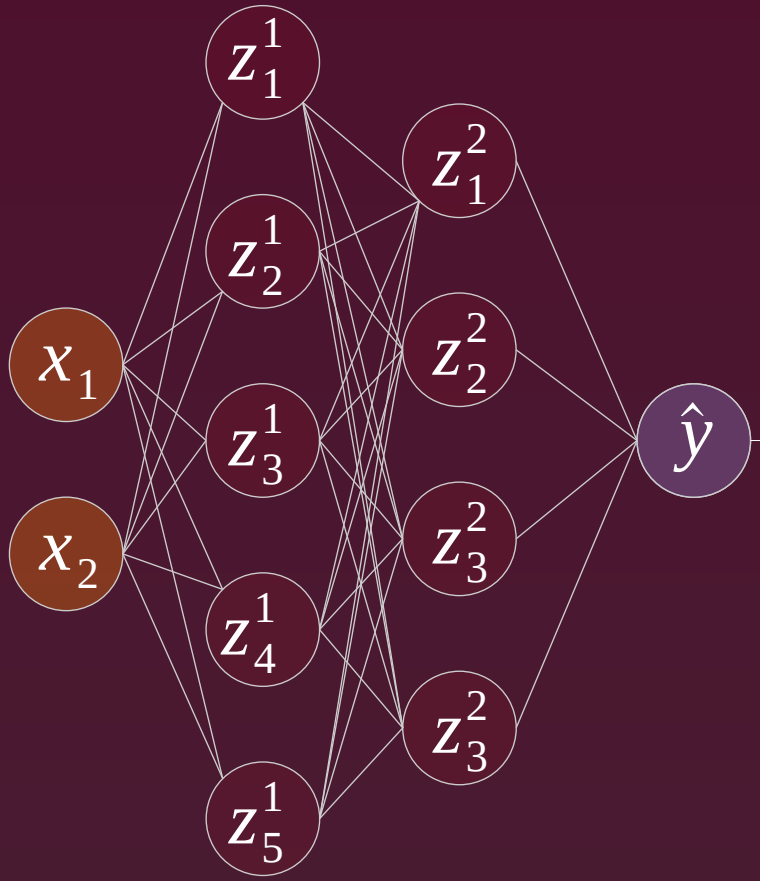
Forward pass



$$\begin{bmatrix} z_1^2 \\ z_2^2 \\ z_3^2 \\ z_4^2 \end{bmatrix} = Relu\left(\begin{bmatrix} w_1^6 & w_2^6 & w_3^6 & w_4^6 & w_5^6 \\ w_1^7 & w_2^7 & w_3^7 & w_4^7 & w_5^7 \\ w_1^8 & w_2^8 & w_3^8 & w_4^8 & w_5^8 \\ w_1^9 & w_2^9 & w_3^9 & w_4^9 & w_5^9 \end{bmatrix} \begin{bmatrix} z_1^1 \\ z_2^1 \\ z_3^1 \\ z_4^1 \\ z_5^1 \end{bmatrix} + \begin{bmatrix} b_1^2 \\ b_2^2 \\ b_3^2 \\ b_4^2 \\ b_5^2 \end{bmatrix} \right)$$

Forward pass

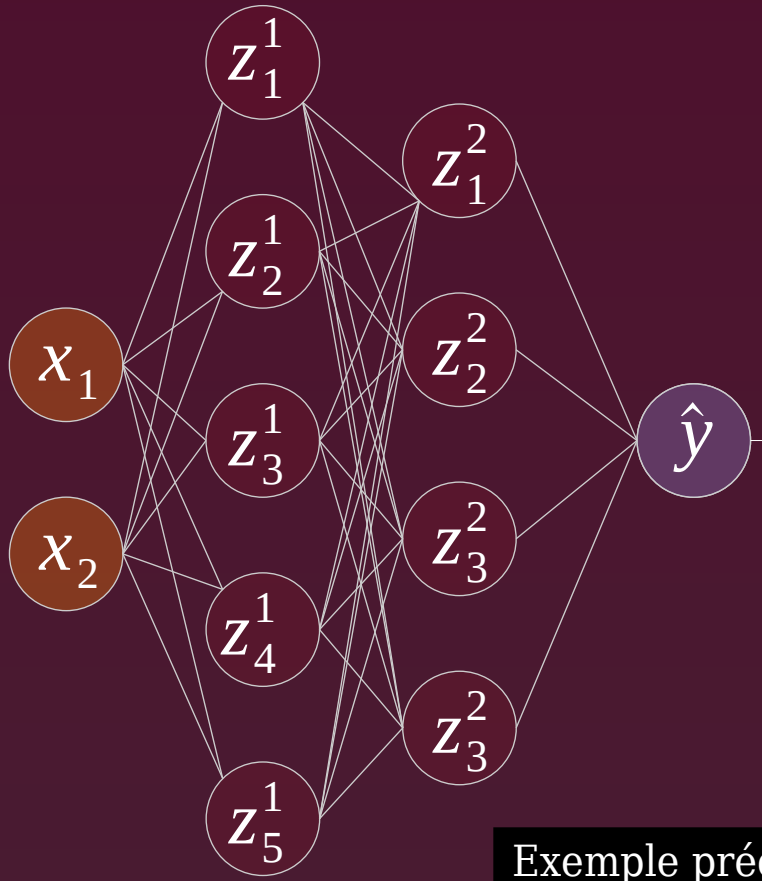
Forward pass



$$\hat{y} = \begin{bmatrix} w_1^{10} & w_2^{10} & w_3^{10} & w_4^{10} \end{bmatrix} \begin{bmatrix} z_1^2 \\ z_2^2 \\ z_3^2 \\ z_4^2 \end{bmatrix} + b^3$$

$$\hat{y} \in \mathbb{R}$$

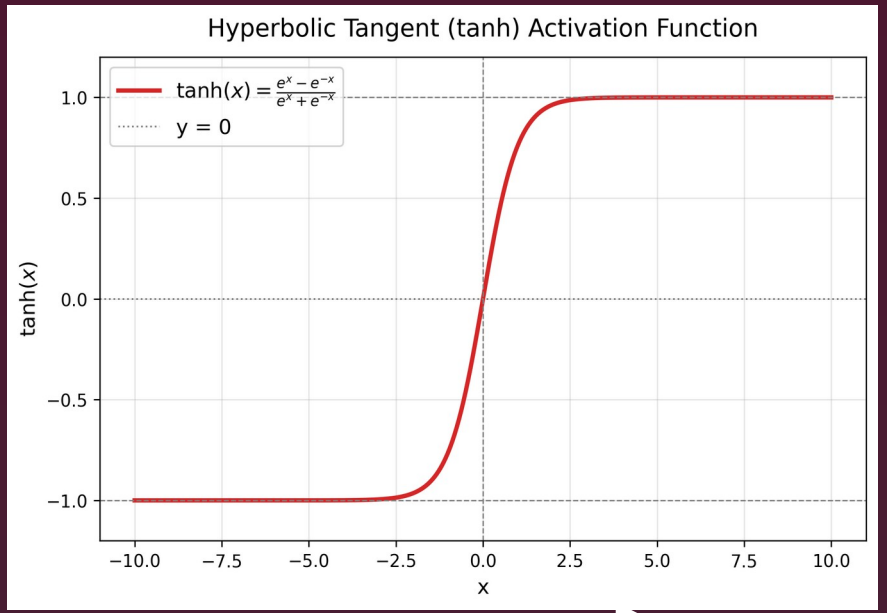
Forward pass



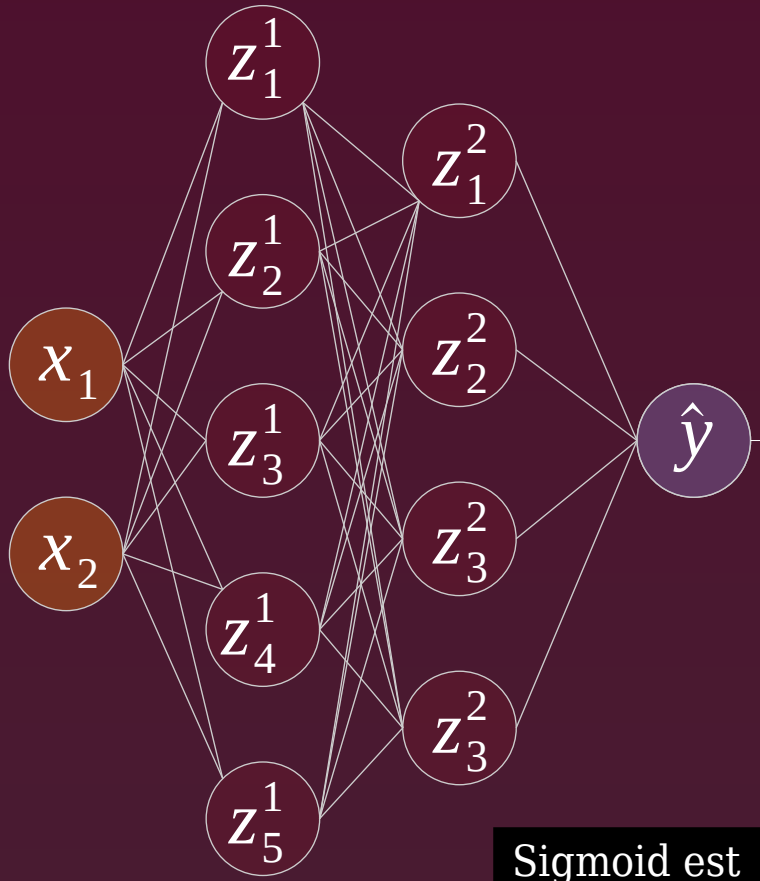
$$\hat{y} = \tanh\left(\begin{bmatrix} w_1^{10} & w_2^{10} & w_3^{10} & w_4^{10} \end{bmatrix} \begin{bmatrix} z_1^2 \\ z_2^2 \\ z_3^2 \\ z_4^2 \end{bmatrix} + b^3\right)$$

$$\hat{y} \in]-1,1[\,;\ \text{car}\ \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Exemple prédire cos(x) ou sin(x).



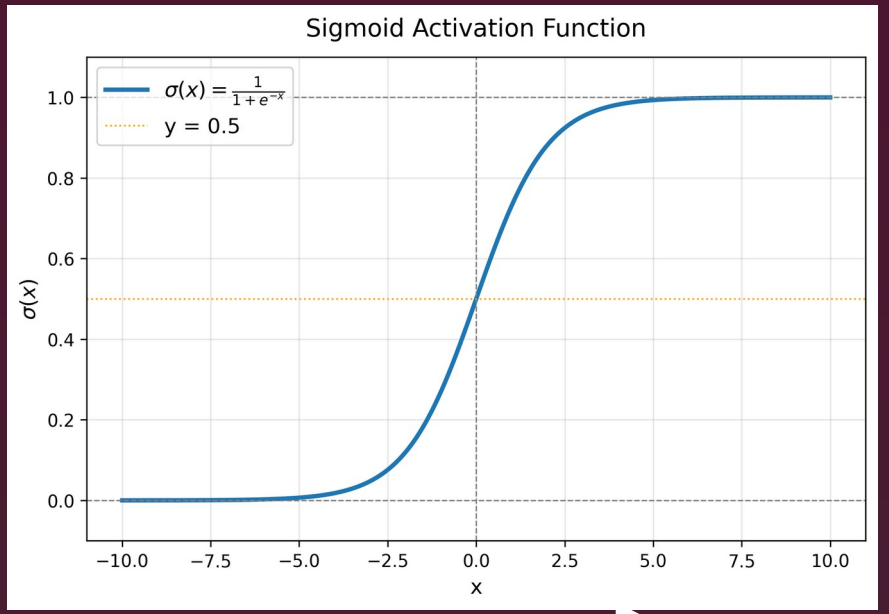
Forward pass



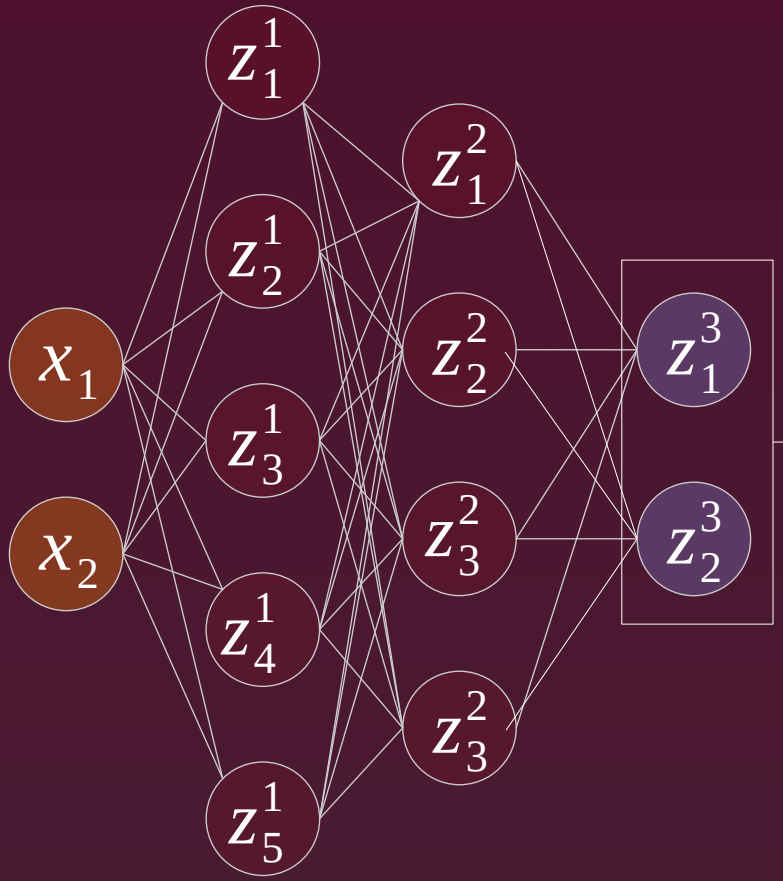
$$\hat{y} = Sigmoid\left(\begin{bmatrix} w_1^{10} & w_2^{10} & w_3^{10} & w_4^{10} \end{bmatrix} \begin{bmatrix} z_1^2 \\ z_2^2 \\ z_3^2 \\ z_4^2 \end{bmatrix} + b^3\right)$$

$$\hat{y} \in]0,1[\ ;\ \text{car}\ \ Sigmoid(x) = \frac{1}{1+e^{-x}}$$

Sigmoid est utilisée pour les classifications binaires.



Forward pass



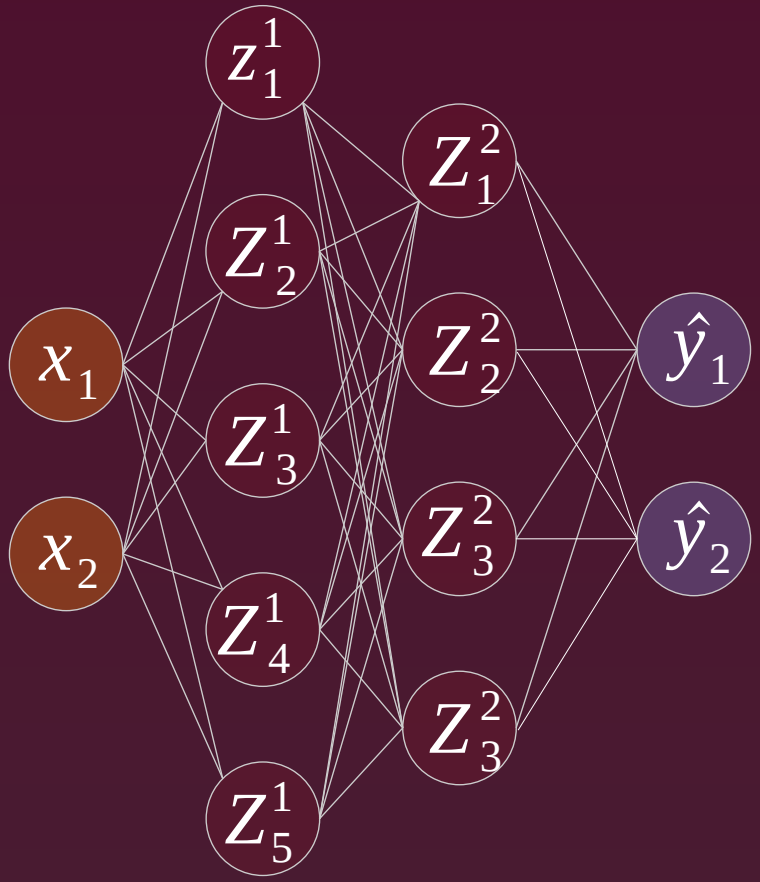
$$\hat{y} = Softmax\left(\begin{bmatrix} z_1^3 \\ z_2^3 \end{bmatrix}\right) = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix}$$

$$\hat{y}_1 = Softmax\left(z_1^3\right) = \frac{e^{z_1^3}}{\sum_{j=1}^{2} e^{z_j^3}}$$

$$\hat{y}_2 = Softmax\left(z_2^3\right) = \frac{e^{z_2^3}}{\sum_{j=1}^{2} e^{z_j^3}}$$

Softmax est utilisée pour les multi classifications.

Implémentaion en python



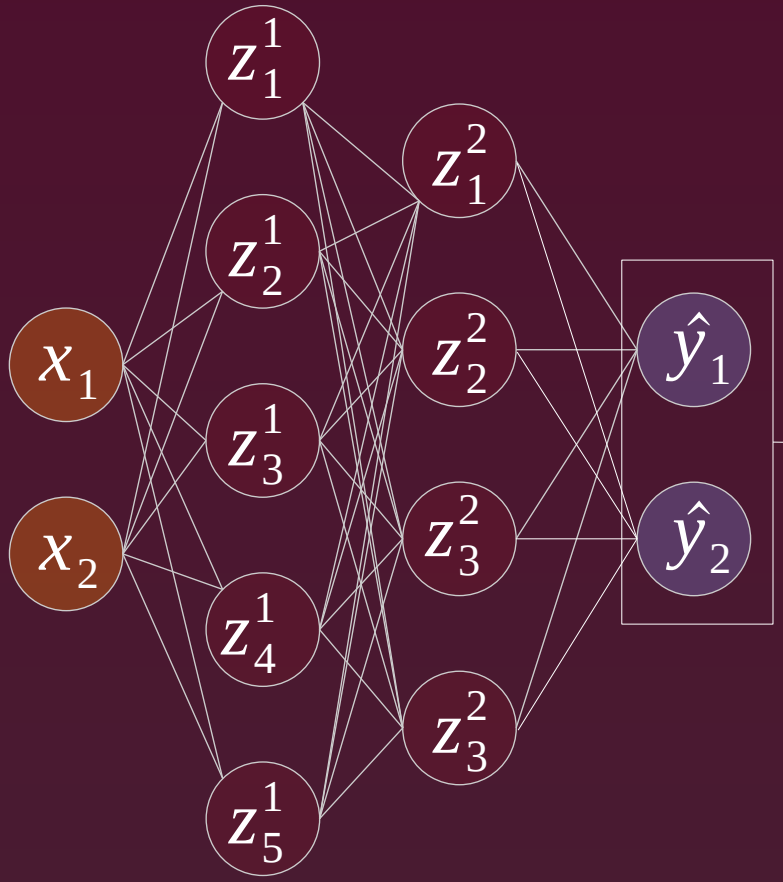
Codons le modèle à l'aide de TensorFlow et PyTorch , deux bibliothèques Python de deep learning.

TensorFlow

PyTorch

- Python is a programming language developed in C;
- Imperative and interpreted language;
- The most widely used language in data science;
- Includes various libraries such as Pandas for handling structured data, NumPy for mathematical calculations, etc.

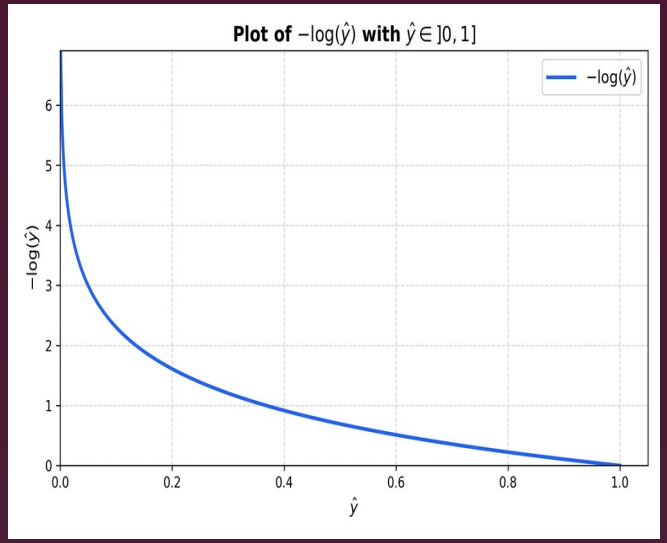
Fonction de perte



$$Erreur = -\sum_{i=1}^{2} y_i \log(\hat{y}_i)$$

Categorical cross-entropy loss

$\hat{y}$ $\qquad$ $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$



Exemple 1

$y = [0,1]^T \quad \hat{y} = [0.7,0.2]^T$

$E = -0\log(0.7) - 1\log(0.2)$

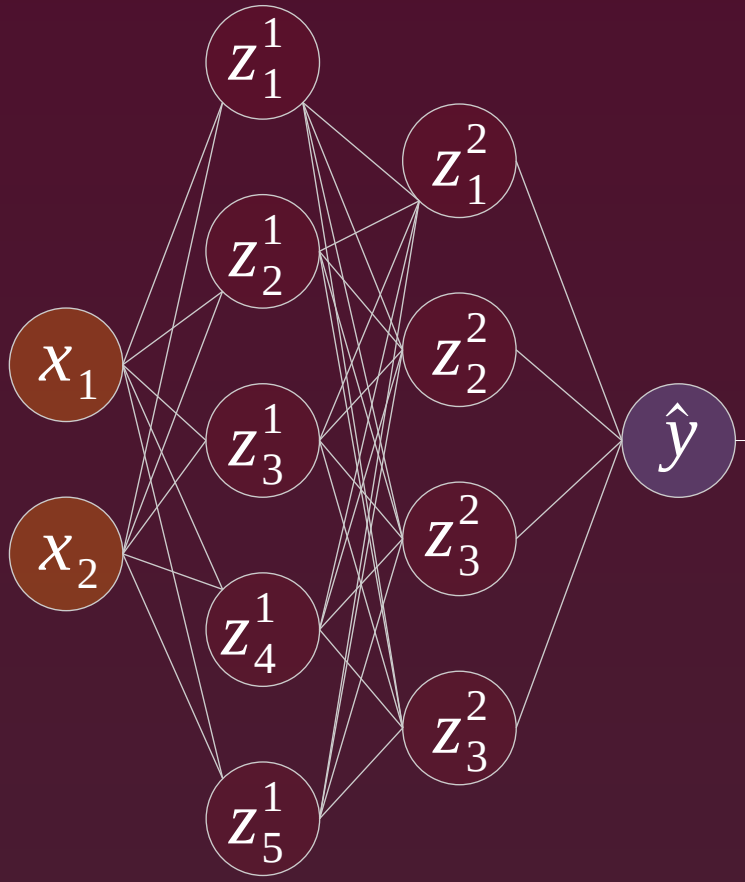
Exemple 2

$y = [1,0]^T \quad \hat{y} = [0.7,0.2]^T$

$E = -1\log(0.7) - 0\log(0.2)$

Loss computation

Fonction de perte

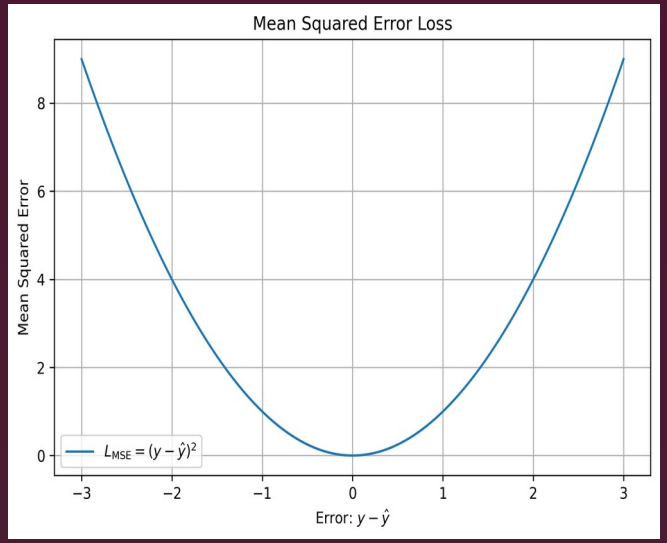


$$Erreur = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

n représente le nombre d'échantillons

Mean Squared Error (MSE) loss

y



Exemple 1

$y = 10$ $\hat{y} = 12$

$E = (10 - 12)^2$

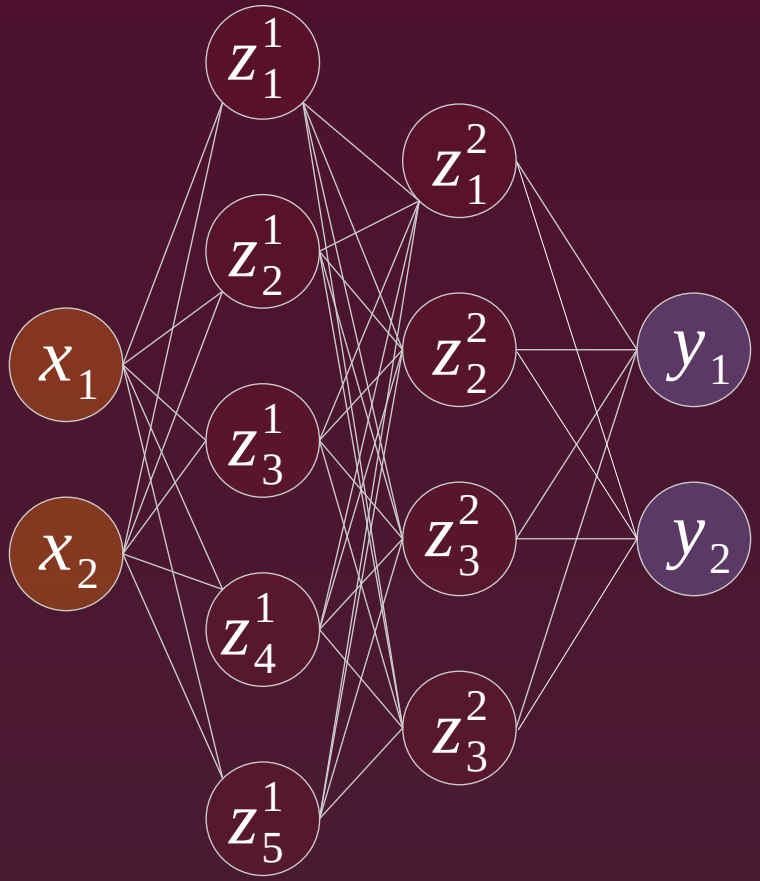
Exemple 2

$y = 2.5$ $\hat{y} = 2$

$E = (2.5 - 2)^2$

Loss computation

Implémentation en python



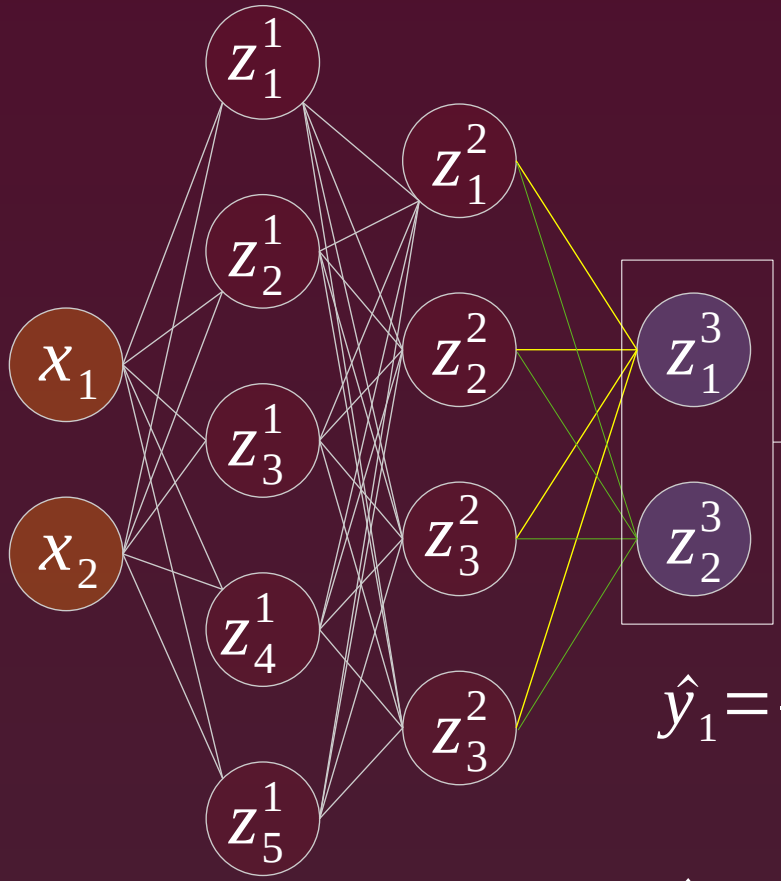
Codons les fonctions de perte à l'aide de TensorFlow et PyTorch , deux bibliothèques Python de deep learning.

TensorFlow

PyTorch

Forward pass

Détails forward pass



$$\mathbf{z^2} = [z_1^2, z_2^2, z_3^2, z_4^2]^T$$

$$Softmax\left(z_1^3\right) = \frac{e^{z_1^3}}{\sum_{j=1}^{2} e^{z_j^3}}$$

$$z^3 = \begin{bmatrix} w_1^1 z_1^2 + w_2^1 z_2^2 + w_3^1 z_3^2 + w_4^1 z_4^2 \\ w_1^2 z_1^2 + w_2^2 z_2^2 + w_3^2 z_3^2 + w_2^2 z_4^2 \end{bmatrix} + b^3 = \begin{bmatrix} z_1^3 \\ z_2^3 \end{bmatrix}$$

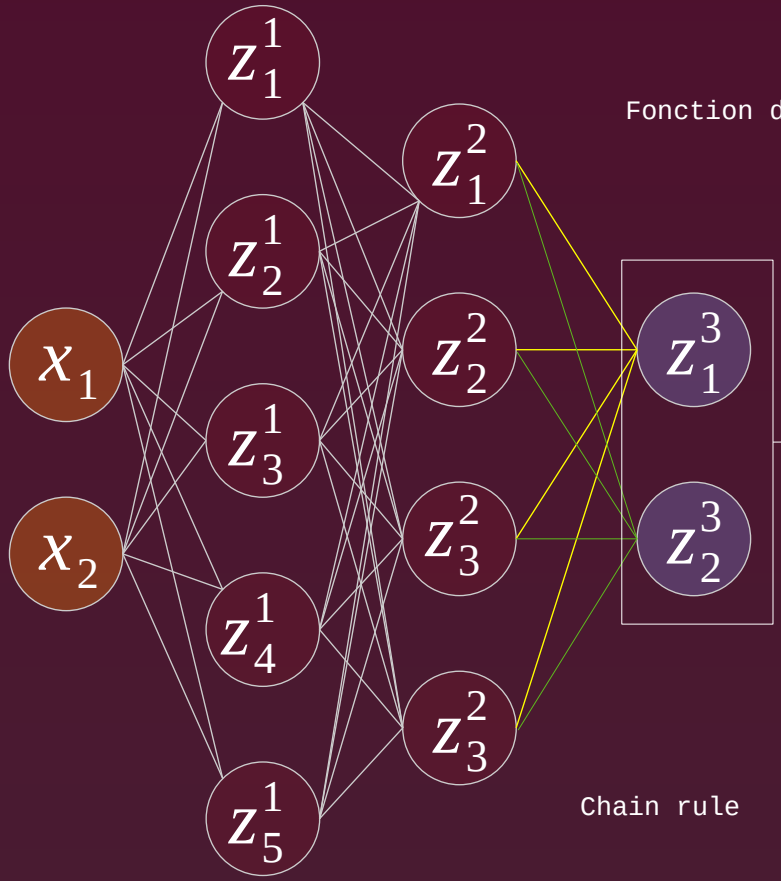
$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix} \quad | \quad \hat{y} = Softmax\left([z_1^3, z_2^3]\right)^T = [\hat{y}_1, \hat{y}_2]^T$$

$$\hat{y}_1 = \frac{e^{z_1^3 = z_1^2 w_1^1 + z_2^2 w_2^1 + z_3^2 w_3^1 + z_4^2 w_4^1 + b_3}}{e^{z_1^3 = z_1^2 w_1^1 + z_2^2 w_2^1 + z_3^2 w_3^1 + z_4^2 w_4^1 + b_3} + e^{z_2^3 = z_1^2 w_1^2 + z_2^2 w_2^2 + z_3^2 w_3^2 + z_4^2 w_4^2 + b_3}}$$

$$\hat{y}_2 = \frac{e^{z_2^3 = z_1^2 w_1^2 + z_2^2 w_2^2 + z_3^2 w_3^2 + z_4^2 w_4^2 + b_3}}{e^{z_1^3 = z_1^2 w_1^1 + z_2^2 w_2^1 + z_3^2 w_3^1 + z_4^2 w_4^1 + b_3} + e^{z_2^3 = z_1^2 w_1^2 + z_2^2 w_2^2 + z_3^2 w_3^2 + z_4^2 w_4^2 + b_3}}$$

Forward pass

Backward pass



Fonction de perte

$$\mathscr{L} = -\sum_{i=1}^{2} y_i \log(\hat{y}_i)$$

Valeur à prédire

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Softmax

$$\hat{\boldsymbol{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix} \quad | \quad \hat{\boldsymbol{y}} = Softmax([z_1^3, z_2^3])^T = [\hat{y}_1, \hat{y}_2]^T$$

Dérivons la fonction de perte en fonction des paramètres

Chain rule

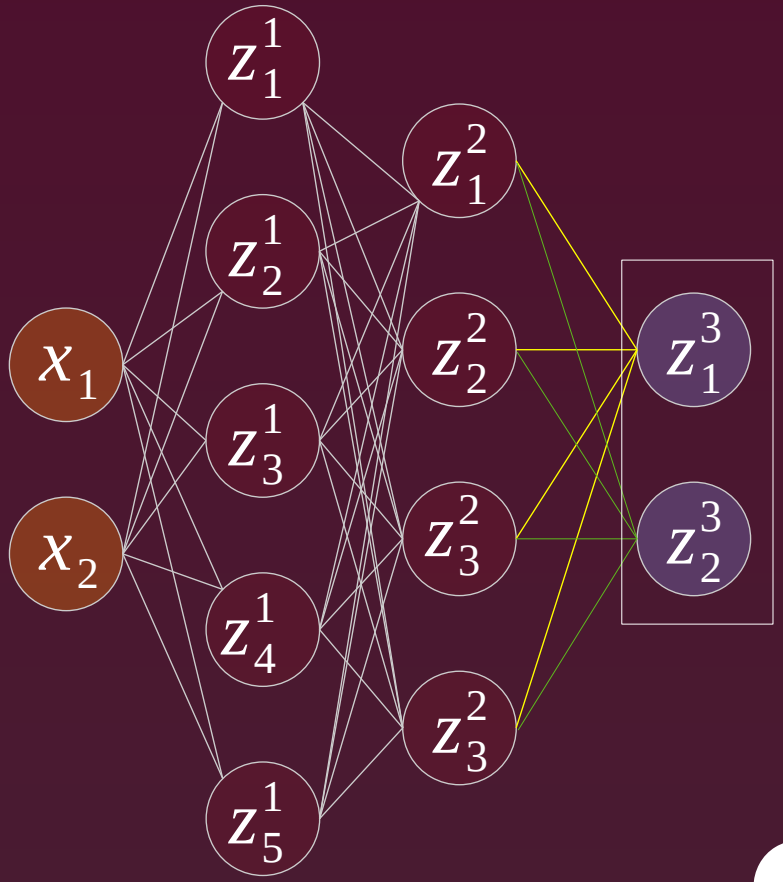
$$\frac{\partial \mathscr{L}}{\partial \boldsymbol{W}} = \frac{\partial \mathscr{L}}{\partial \hat{\boldsymbol{y}}} \frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{z^3}} \frac{\partial \boldsymbol{z^3}}{\partial \boldsymbol{W}}$$

$$| \quad \boldsymbol{W} = \begin{bmatrix} w_1^1 & w_2^1 & w_3^1 & w_4^1 \\ w_1^2 & w_2^2 & w_3^2 & w_2^2 \end{bmatrix}$$

$$| \quad \boldsymbol{z^3} = [z_1^3 z_2^3]^T$$

Ce terme n'est pas théoriquement correct; nous y reviendrons.

Backward pass



Chaîne rule

$$\frac{\partial \mathscr{L}}{\partial W} = \frac{\partial \mathscr{L}}{\partial \hat{y}^T} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\mathscr{L} = -\sum_{i=1}^{2} y_i \log(\hat{y}_i) = -(y_1 \log(\hat{y}_1) + y_2 \log(\hat{y}_2))$$

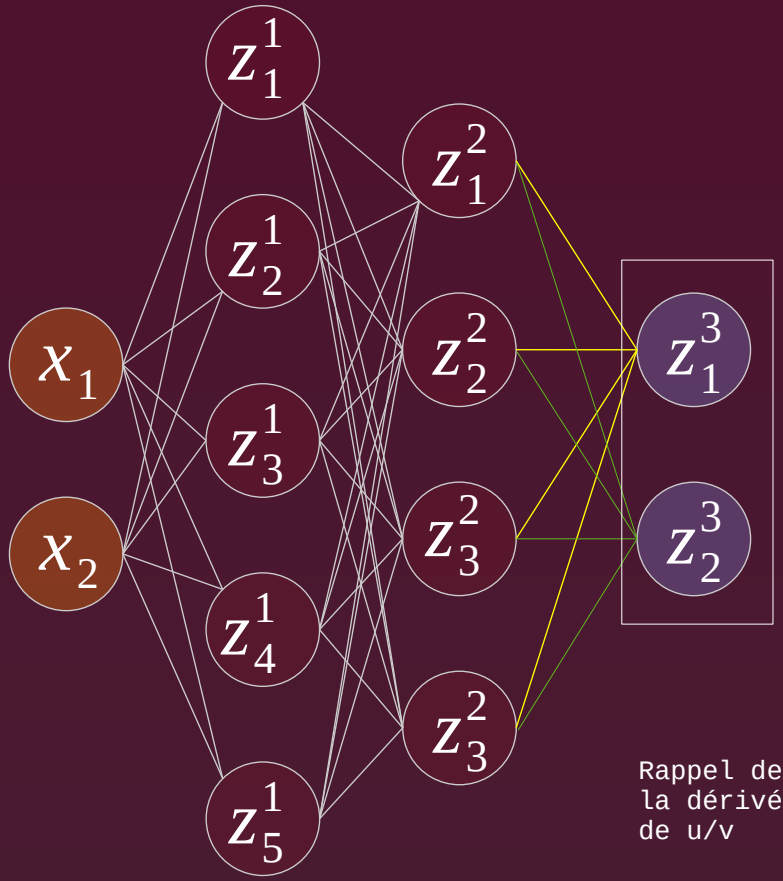
$$\mathscr{L} = -(y_1 \log(\hat{y}_1) + y_2 \log(\hat{y}_2)) = -\begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} \log(\hat{y}_1) \\ \log(\hat{y}_2) \end{bmatrix}$$

$$\mathscr{L} = -\begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} \log(\hat{y}_1) \\ \log(\hat{y}_2) \end{bmatrix} = -\begin{bmatrix} y_1 & y_2 \end{bmatrix} \log\left(\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix}\right)$$

1

$$\frac{\partial \mathscr{L}}{\partial \hat{y}} = \begin{bmatrix} \frac{\partial \mathscr{L}}{\partial \hat{y}_1} \\ \frac{\partial \mathscr{L}}{\partial \hat{y}_2} \end{bmatrix} = \begin{bmatrix} \frac{-y_1}{\hat{y}_1} \\ \frac{-y_2}{\hat{y}_2} \end{bmatrix}; \quad \text{car}(\ln(x))' = \frac{1}{x}$$

Backward pass



Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\hat{y} = [\hat{y}_1 \hat{y}_2]^T ;\ \hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} ;\ \hat{y}_2 = \frac{e^{z_2^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} ;\ z^3 = [z_1^3 z_2^3]^T$$

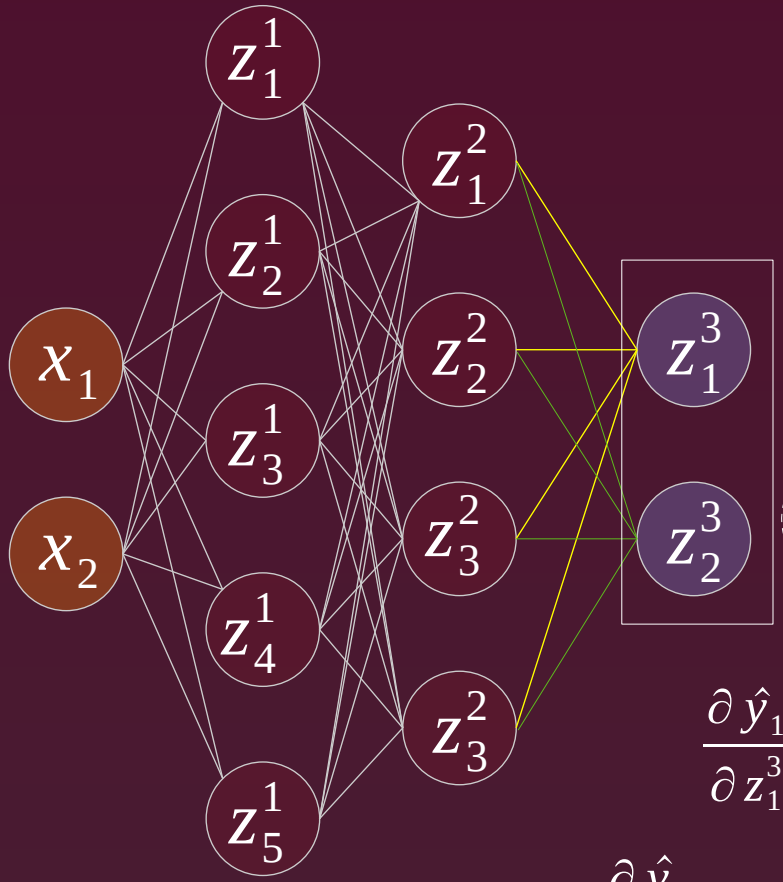
$$\hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} = \frac{e^{b_3} e^{z_1^3}}{e^{b_3}\left(e^{z_1^3} + e^{z_2^3}\right)} = \frac{e^{z_1^3}}{e^{z_1^3} + e^{z_2^3}} ;\ \hat{y}_2 = \frac{e^{z_2^3}}{e^{z_1^3} + e^{z_2^3}}$$

Rappel de la dérivé de u/v

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

$$\frac{\partial \hat{y}}{\partial z^3} = \begin{bmatrix} \frac{\partial \hat{y}_1}{\partial z_1^3} & \frac{\partial \hat{y}_1}{\partial z_2^3} \\ \frac{\partial \hat{y}_2}{\partial z_1^3} & \frac{\partial \hat{y}_2}{\partial z_2^3} \end{bmatrix}$$

Backward pass



Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} = \frac{e^{b_3} e^{z_1^3}}{e^{b_3}\left(e^{z_1^3} + e^{z_2^3}\right)} = \frac{e^{z_1^3}}{e^{z_1^3} + e^{z_2^3}};\ \hat{y}_2 = \frac{e^{z_2^3}}{e^{z_1^3} + e^{z_2^3}}$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} \mid u = e^{z_1^3};\ v = e^{z_1^3} + e^{z_2^3} \rightarrow u' = e^{z_1^3};\ v' = e^{z_1^3}$$

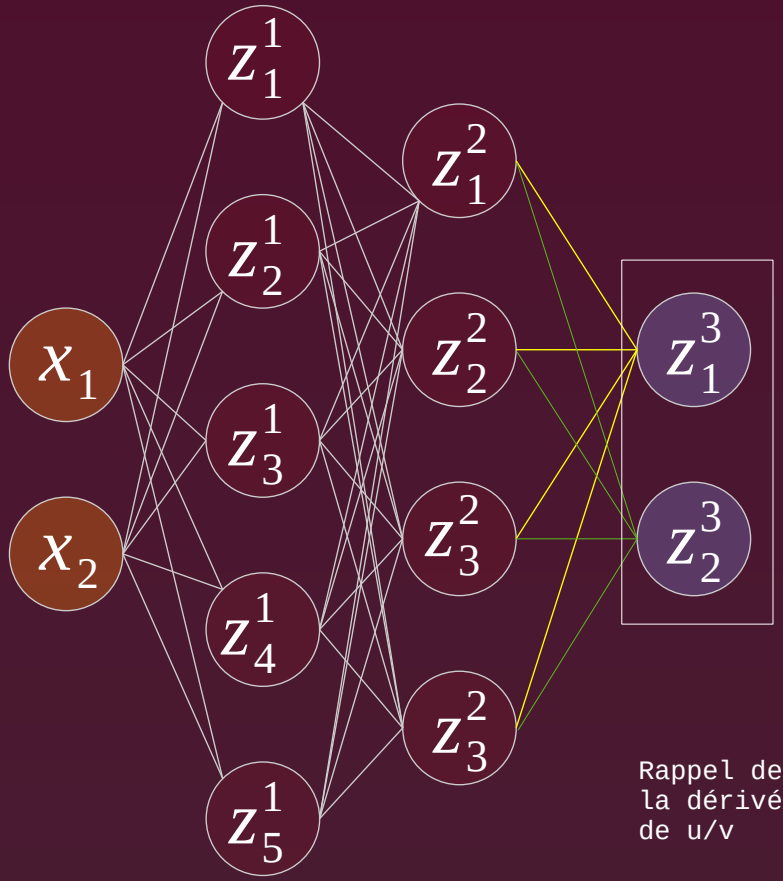
Dérivons $\hat{y}_1$, U et v en fonction de z^3_1

$$\frac{\partial \hat{y}_1}{\partial z_1^3} = \frac{e^{z_1^3}\left(e^{z_1^3} + e^{z_2^3}\right) - e^{z_1^3}\left(e^{z_1^3}\right)}{\left(e^{z_1^3} + e^{z_2^3}\right)^2}$$

$$\frac{\partial \hat{y}_1}{\partial z_1^3} = \frac{e^{z_1^3}\left(e^{z_1^3} + e^{z_2^3}\right) - e^{z_1^3}\left(e^{z_1^3}\right)}{\left(e^{z_1^3} + e^{z_2^3}\right)^2} = \frac{e^{z_1^3} e^{z_1^3} + e^{z_1^3} e^{z_2^3} - e^{z_1^3} e^{z_1^3}}{\left(e^{z_1^3} + e^{z_2^3}\right)^2} = \frac{e^{z_1^3} e^{z_2^3}}{\left(e^{z_1^3} + e^{z_2^3}\right)^2}$$

$$\frac{\partial \hat{y}_1}{\partial z_1^3} = \frac{e^{z_1^3} e^{z_2^3}}{\left(e^{z_1^3} + e^{z_2^3}\right)^2} = \frac{e^{z_1^3}}{e^{z_1^3} + e^{z_2^3}} \frac{e^{z_2^3}}{e^{z_1^3} + e^{z_2^3}} = \hat{y}_1 \hat{y}_2 = \hat{y}_1\left(1 - \hat{y}_1\right);\ \text{car}\ \ \hat{y}_1 + \hat{y}_2 = 1$$

Backward pass



Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

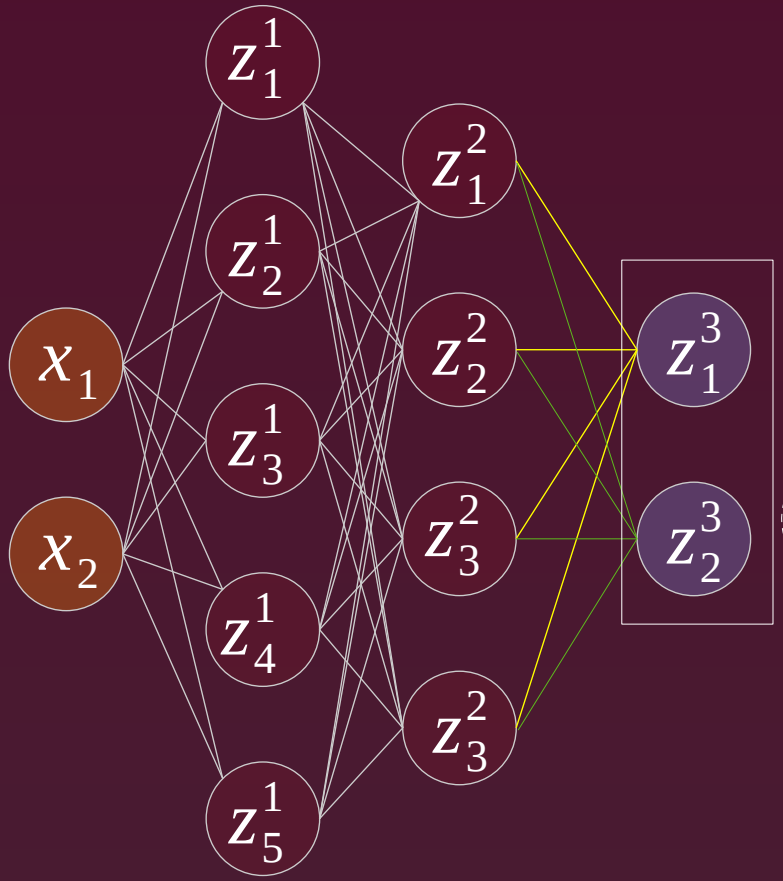
$$\hat{y} = [\hat{y}_1 \hat{y}_2]^T ;\ \hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} ;\ \hat{y}_2 = \frac{e^{z_2^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} ;\ z^3 = [z_1^3 z_2^3]^T$$

$$\hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} = \frac{e^{b_3} e^{z_1^3}}{e^{b_3}\left(e^{z_1^3} + e^{z_2^3}\right)} = \frac{e^{z_1^3}}{e^{z_1^3} + e^{z_2^3}} ;\ \hat{y}_2 = \frac{e^{z_2^3}}{e^{z_1^3} + e^{z_2^3}}$$

Rappel de la dérivé de u/v

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

$$\frac{\partial \hat{y}}{\partial z^3} = \begin{bmatrix} \frac{\partial \hat{y}_1}{\partial z_1^3} & \frac{\partial \hat{y}_1}{\partial z_2^3} \\ \frac{\partial \hat{y}_2}{\partial z_1^3} & \frac{\partial \hat{y}_2}{\partial z_2^3} \end{bmatrix}$$



Backward pass

Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} = \frac{e^{b_3} e^{z_1^3}}{e^{b_3}\left(e^{z_1^3} + e^{z_2^3}\right)} = \frac{e^{z_1^3}}{e^{z_1^3} + e^{z_2^3}}; \quad \hat{y}_2 = \frac{e^{z_2^3}}{e^{z_1^3} + e^{z_2^3}}$$

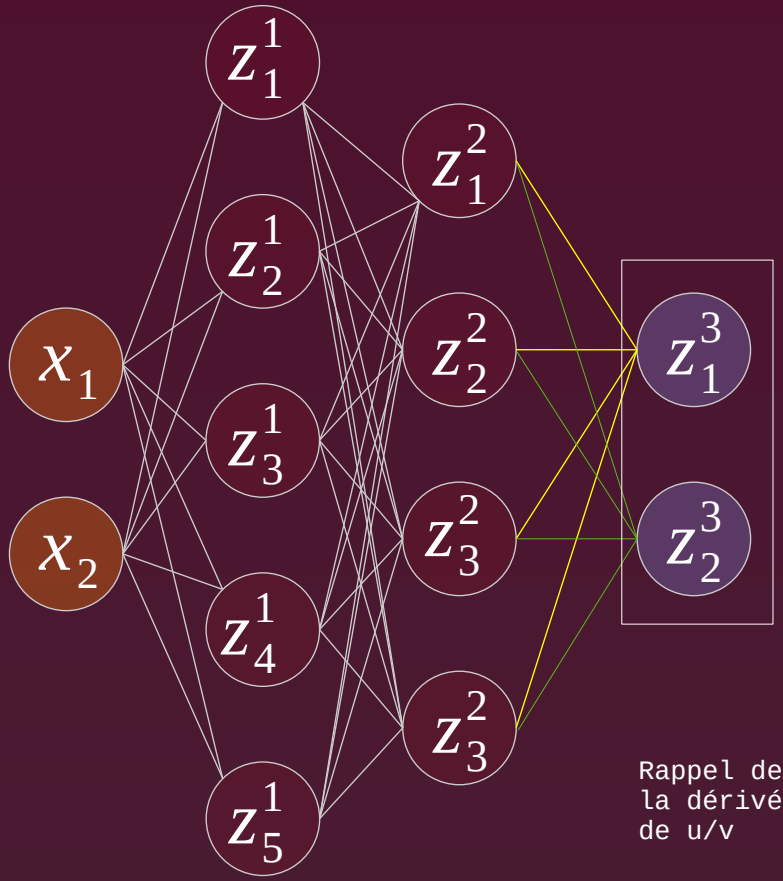
$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} \mid u = e^{z_1^3};\ v = e^{z_1^3} + e^{z_2^3} \rightarrow u' = 0;\ v' = e^{z_2^3}$$

Dérivons $\hat{y}_1$, U et v en fonction de z^3_2

$$\frac{\partial \hat{y}_1}{\partial z_2^3} = \frac{0\left(e^{z_1^3} + e^{z_2^3}\right) - e^{z_1^3}\left(e^{z_2^3}\right)}{\left(e^{z_1^3} + e^{z_2^3}\right)^2}$$

$$\frac{\partial \hat{y}_1}{\partial z_1^3} = \frac{-e^{z_1^3}\left(e^{z_2^3}\right)}{\left(e^{z_1^3} + e^{z_2^3}\right)^2} = \frac{-e^{z_1^3}}{\left(e^{z_1^3} + e^{z_2^3}\right)} \frac{e^{z_2^3}}{\left(e^{z_1^3} + e^{z_2^3}\right)} = -\hat{y}_1 \hat{y}_2$$

Backward pass



Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\hat{y} = [\hat{y}_1 \hat{y}_2]^T ;\ \hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} ;\ \hat{y}_2 = \frac{e^{z_2^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} ;\ z^3 = [z_1^3 z_2^3]^T$$

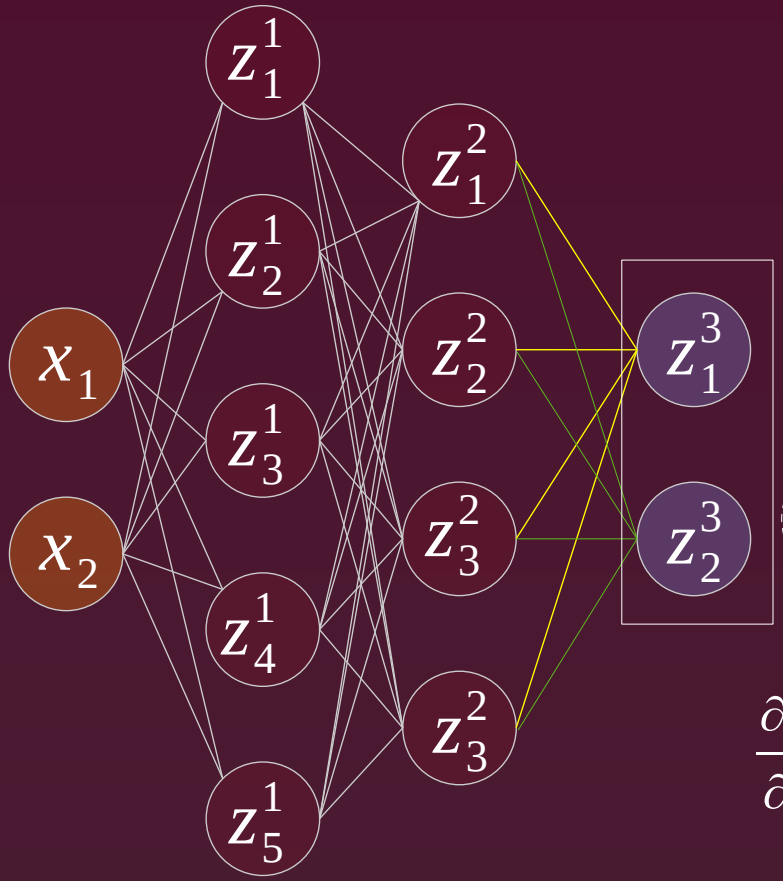
$$\hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} = \frac{e^{b_3} e^{z_1^3}}{e^{b_3}\left(e^{z_1^3} + e^{z_2^3}\right)} = \frac{e^{z_1^3}}{e^{z_1^3} + e^{z_2^3}} ;\ \hat{y}_2 = \frac{e^{z_2^3}}{e^{z_1^3} + e^{z_2^3}}$$

Rappel de la dérivé de u/v

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Backward pass



Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} = \frac{e^{b_3} e^{z_1^3}}{e^{b_3}\left(e^{z_1^3} + e^{z_2^3}\right)} = \frac{e^{z_1^3}}{e^{z_1^3} + e^{z_2^3}}; \quad \hat{y}_2 = \frac{e^{z_2^3}}{e^{z_1^3} + e^{z_2^3}}$$

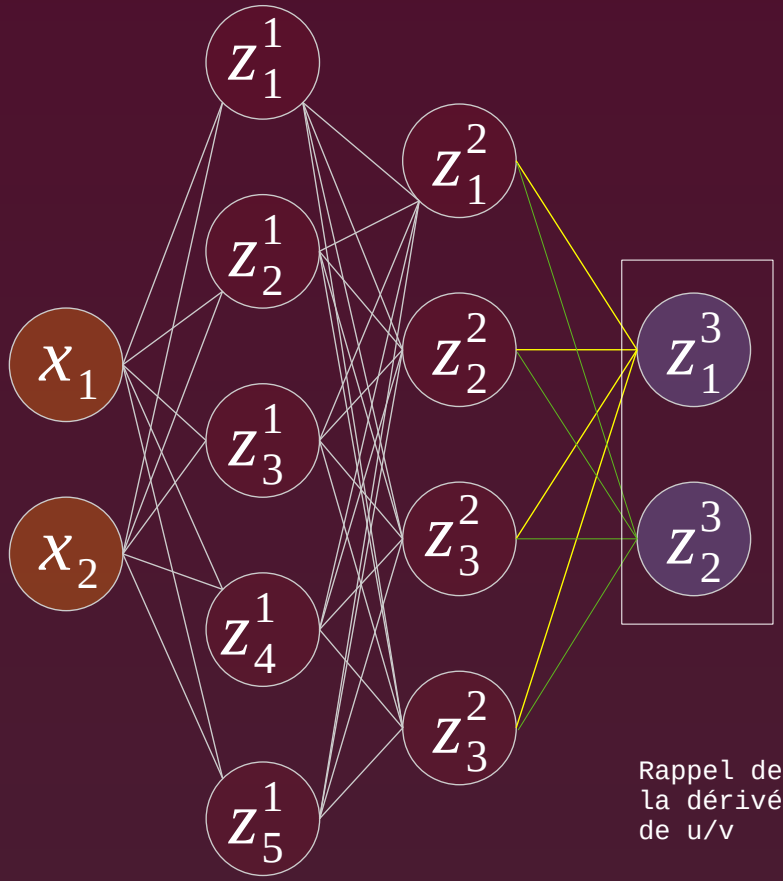
$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} \mid u = e^{z_2^3};\ v = e^{z_1^3} + e^{z_2^3} \rightarrow u' = 0;\ v' = e^{z_1^3}$$

Dérivons $\hat{y}_2$, U et v en fonction de z^3_1

$$\frac{\partial \hat{y}_2}{\partial z_1^3} = \frac{0\left(e^{z_1^3} + e^{z_2^3}\right) - e^{z_2^3}\left(e^{z_1^3}\right)}{\left(e^{z_1^3} + e^{z_2^3}\right)^2}$$

$$\frac{\partial \hat{y}_2}{\partial z_1^3} = \frac{-e^{z_2^3}\left(e^{z_1^3}\right)}{\left(e^{z_1^3} + e^{z_2^3}\right)^2} = \frac{-e^{z_2^3}}{\left(e^{z_1^3} + e^{z_2^3}\right)} \frac{e^{z_1^3}}{\left(e^{z_1^3} + e^{z_2^3}\right)} = -\hat{y}_2 \hat{y}_1; \text{ car } \hat{y}_1 + \hat{y}_2 = 1$$

Backward pass



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$$\hat{y} = [\hat{y}_1 \hat{y}_2]^T ;\ \hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} ;\ \hat{y}_2 = \frac{e^{z_2^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} ;\ z^3 = [z_1^3 z_2^3]^T$$

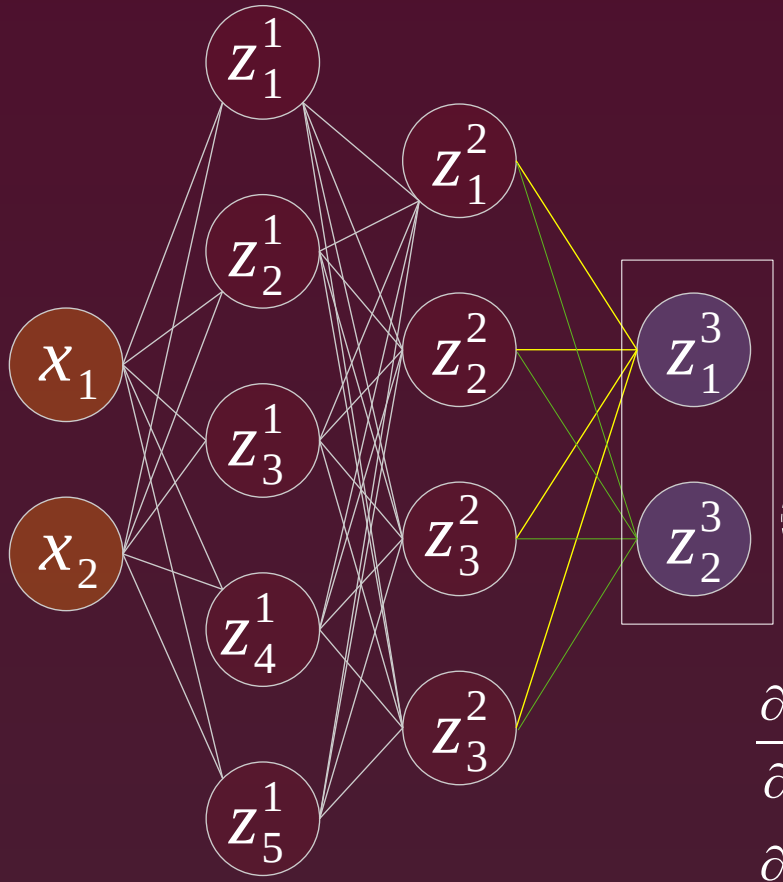
$$\hat{y}_1 = \frac{e^{z_1^3 + b_3}}{e^{z_1^3 + b_3} + e^{z_2^3 + b_3}} = \frac{e^{b_3} e^{z_1^3}}{e^{b_3}\left(e^{z_1^3} + e^{z_2^3}\right)} = \frac{e^{z_1^3}}{e^{z_1^3} + e^{z_2^3}} ;\ \hat{y}_2 = \frac{e^{z_2^3}}{e^{z_1^3} + e^{z_2^3}}$$

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Backward pass



Backward pass

Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\hat{y}_1 = \frac{e^{z^3_1 + b_3}}{e^{z^3_1 + b_3} + e^{z^3_2 + b_3}} = \frac{e^{b_3} e^{z^3_1}}{e^{b_3}\left(e^{z^3_1} + e^{z^3_2}\right)} = \frac{e^{z^3_1}}{e^{z^3_1} + e^{z^3_2}}; \quad \hat{y}_2 = \frac{e^{z^3_2}}{e^{z^3_1} + e^{z^3_2}}$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} \mid u = e^{z^3_2};\ v = e^{z^3_1} + e^{z^3_2} \qquad u' = e^{z^3_2};\ v' = e^{z^3_2}$$

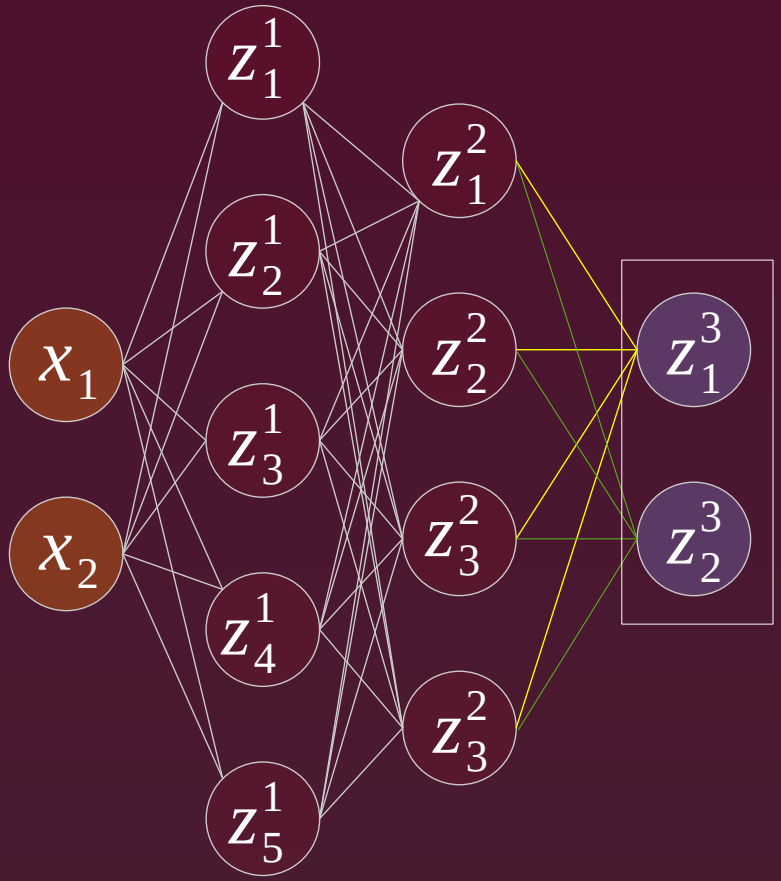
Dérivons $\hat{y}_2$, U et v en fonction de z^3_2

$$\frac{\partial \hat{y}_2}{\partial z^3_2} = \frac{e^{z^3_2}\left(e^{z^3_1} + e^{z^3_2}\right) - e^{z^3_2}\left(e^{z^3_2}\right)}{\left(e^{z^3_1} + e^{z^3_2}\right)^2}$$

$$\frac{\partial \hat{y}_2}{\partial z^3_2} = \frac{e^{z^3_2} e^{z^3_1} + e^{z^3_2} e^{z^3_2} - e^{z^3_2} e^{z^3_2}}{\left(e^{z^3_1} + e^{z^3_2}\right)^2} = \frac{e^{z^3_2} e^{z^3_1}}{\left(e^{z^3_1} + e^{z^3_2}\right)^2} = \frac{e^{z^3_1}}{\left(e^{z^3_1} + e^{z^3_2}\right)} \frac{e^{z^3_2}}{\left(e^{z^3_1} + e^{z^3_2}\right)}$$

$$\frac{\partial \hat{y}_2}{\partial z^3_2} = \hat{y}_1 \hat{y}_2 = \hat{y}_2(1 - \hat{y}_2); \text{ car } \hat{y}_1 + \hat{y}_2 = 1$$

Backward pass



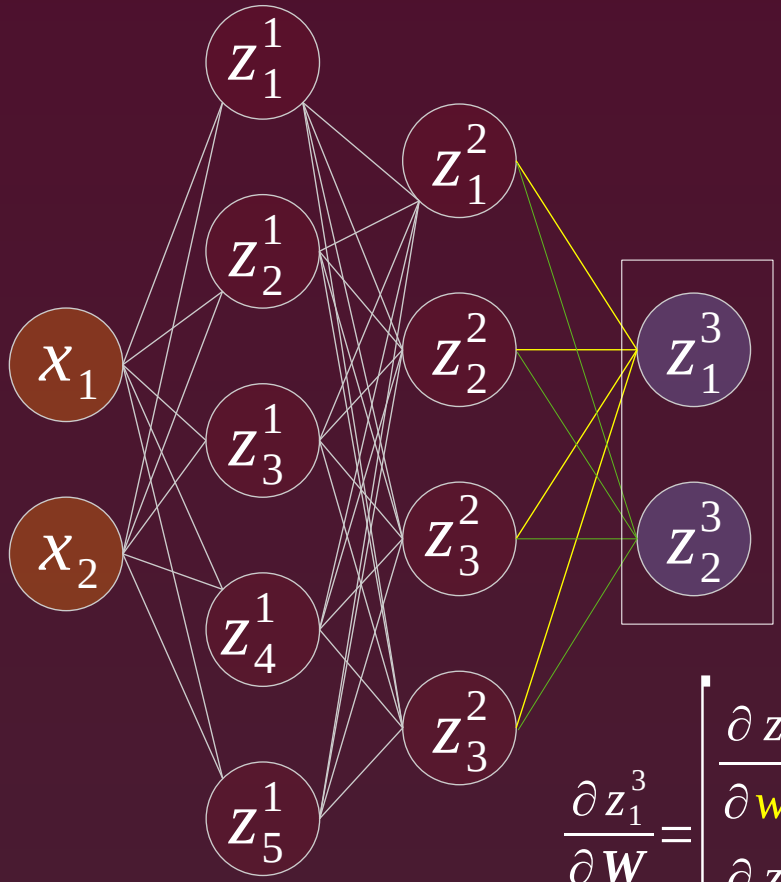
Chaîne rule

$$\frac{\partial \mathscr{L}}{\partial W} = \frac{\partial \mathscr{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\frac{\partial \hat{y}}{\partial z^3} = \begin{bmatrix} \frac{\partial \hat{y}_1}{\partial z_1^3} & \frac{\partial \hat{y}_1}{\partial z_2^3} \\ \frac{\partial \hat{y}_2}{\partial z_1^3} & \frac{\partial \hat{y}_2}{\partial z_2^3} \end{bmatrix} = \begin{bmatrix} \hat{y}_1(1-\hat{y}_1) & -\hat{y}_1\hat{y}_2 \\ -\hat{y}_2\hat{y}_1 & \hat{y}_2(1-\hat{y}_2) \end{bmatrix}$$

2

Backward pass



Chaîne rule

$$\frac{\partial \mathscr{L}}{\partial W} = \frac{\partial \mathscr{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \boxed{\frac{\partial z^3}{\partial W}}$$

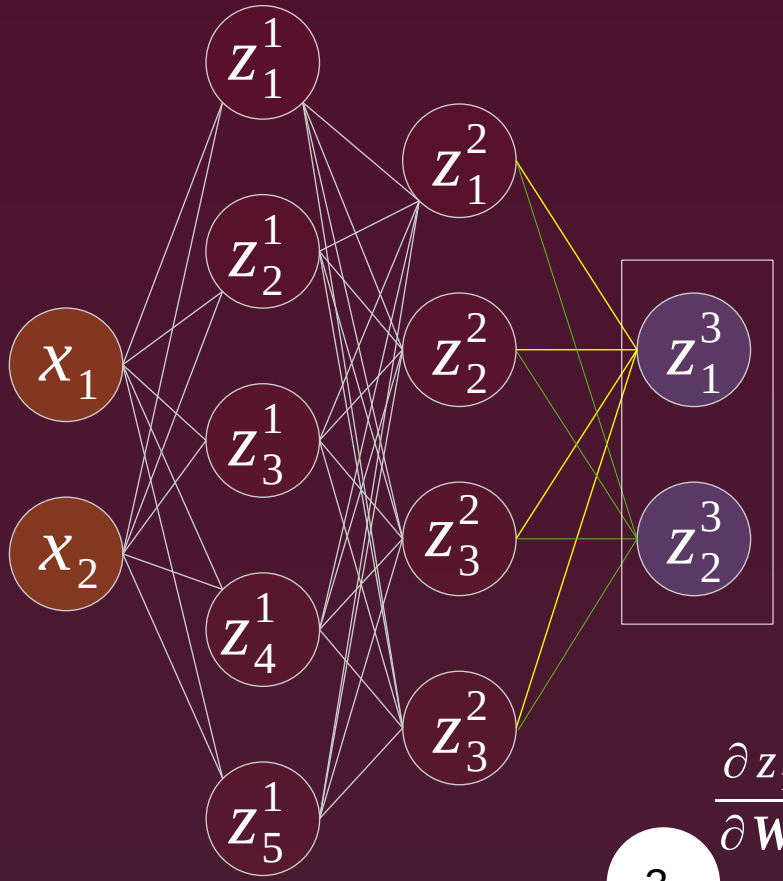
$$Z^3 = W Z^2 + b^3 = \begin{bmatrix} w_1^1 & w_2^1 & w_3^1 & w_4^1 \\ w_1^2 & w_2^2 & w_3^2 & w_2^2 \end{bmatrix} \begin{bmatrix} z_1^2 \\ z_2^2 \\ z_3^2 \\ z_4^2 \end{bmatrix} + b^3$$

$$Z^3 = \begin{bmatrix} w_1^1 z_1^2 + w_2^1 z_2^2 + w_3^1 z_3^2 + w_4^1 z_4^2 \\ w_1^2 z_1^2 + w_2^2 z_2^2 + w_3^2 z_3^2 + w_2^2 z_4^2 \end{bmatrix} = \begin{bmatrix} z_1^3 \\ z_2^3 \end{bmatrix}$$

Ce biais peut être omis.

$$\frac{\partial z_1^3}{\partial W} = \begin{bmatrix} \frac{\partial z_1^3}{\partial w_1^1} & \frac{\partial z_1^3}{\partial w_2^1} & \frac{\partial z_1^3}{\partial w_3^1} & \frac{\partial z_1^3}{\partial w_4^1} \\ \frac{\partial z_1^3}{\partial w_1^2} & \frac{\partial z_1^3}{\partial w_2^2} & \frac{\partial z_1^3}{\partial w_3^2} & \frac{\partial z_1^3}{\partial w_4^2} \end{bmatrix} \qquad \frac{\partial z_2^3}{\partial W} = \begin{bmatrix} \frac{\partial z_2^3}{\partial w_1^1} & \frac{\partial z_2^3}{\partial w_2^1} & \frac{\partial z_2^3}{\partial w_3^1} & \frac{\partial z_2^3}{\partial w_4^1} \\ \frac{\partial z_2^3}{\partial w_1^2} & \frac{\partial z_2^3}{\partial w_2^2} & \frac{\partial z_2^3}{\partial w_3^2} & \frac{\partial z_2^3}{\partial w_4^2} \end{bmatrix}$$

Backward pass



Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \boxed{\frac{\partial z^3}{\partial W}}$$

$$z^3 = W z^2 = \begin{bmatrix} w_1^1 & w_2^1 & w_3^1 & w_4^1 \\ w_1^2 & w_2^2 & w_3^2 & w_2^2 \end{bmatrix} \begin{bmatrix} z_1^2 \\ z_2^2 \\ z_3^2 \\ z_4^2 \end{bmatrix}$$

$$z^3 = \begin{bmatrix} w_1^1 z_1^2 + w_2^1 z_2^2 + w_3^1 z_3^2 + w_4^1 z_4^2 \\ w_1^2 z_1^2 + w_2^2 z_2^2 + w_3^2 z_3^2 + w_2^2 z_4^2 \end{bmatrix} = \begin{bmatrix} z_1^3 \\ z_2^3 \end{bmatrix}$$

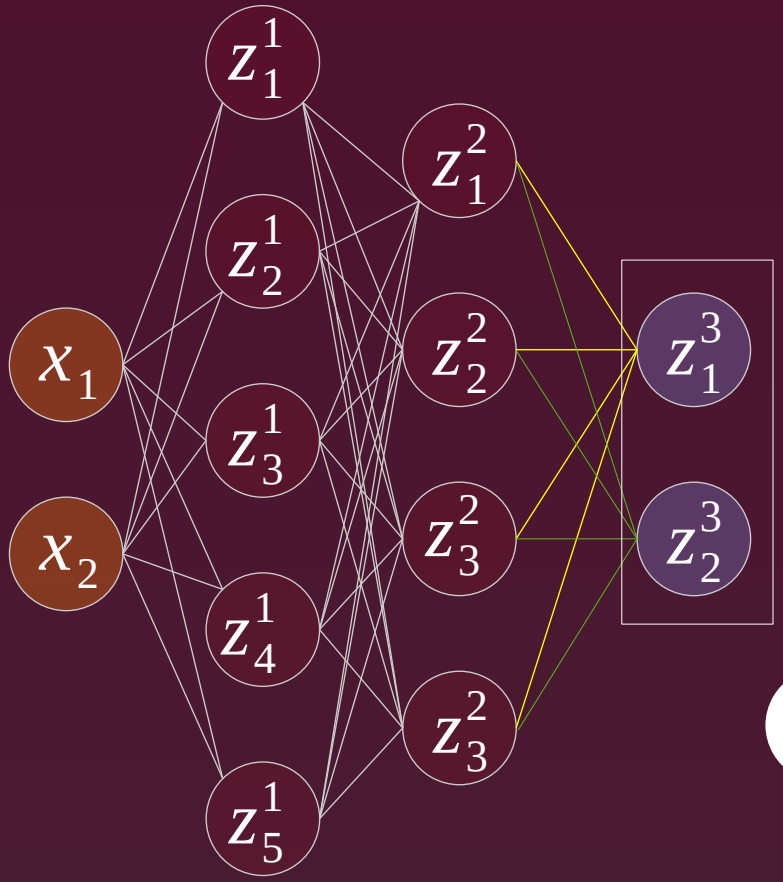
3_1

$$\frac{\partial z_1^3}{\partial W} = \begin{bmatrix} z_1^2 & z_2^2 & z_3^2 & z_4^2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3_2

$$\frac{\partial z_2^3}{\partial W} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ z_1^2 & z_2^2 & z_3^2 & z_4^2 \end{bmatrix}$$

Backward pass



Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = \begin{bmatrix} \frac{-y_1}{\hat{y}_1} \\ \frac{-y_2}{\hat{y}_2} \end{bmatrix}$$

$$\frac{\partial \hat{y}}{\partial z^3} = \begin{bmatrix} \hat{y}_1(1-\hat{y}_1) & -\hat{y}_1\hat{y}_2 \\ -\hat{y}_2\hat{y}_1 & \hat{y}_2(1-\hat{y}_2) \end{bmatrix}$$

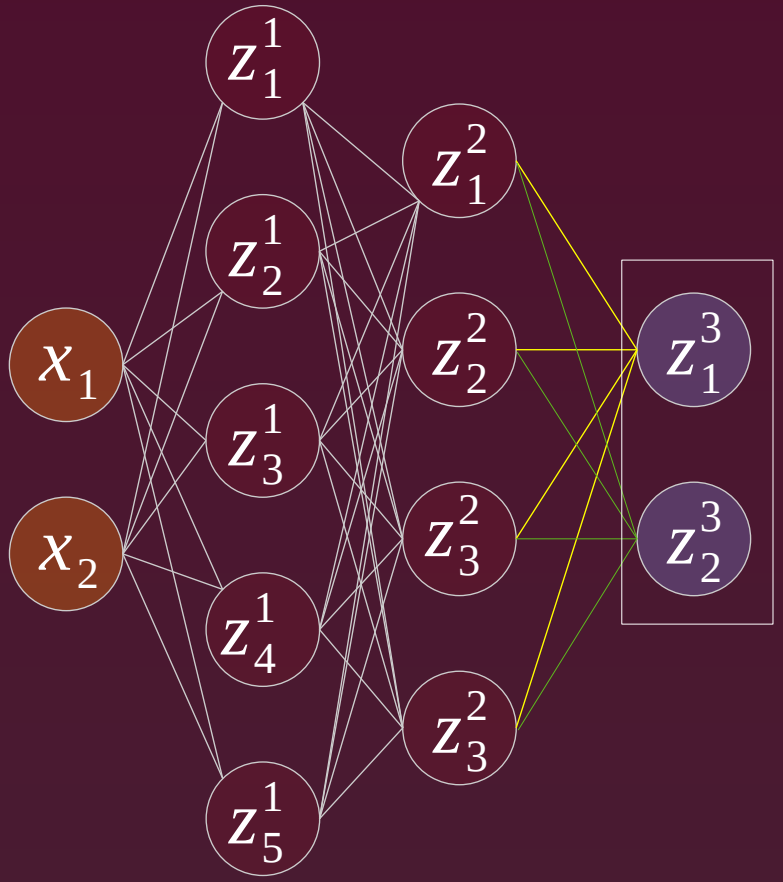
1 2

$$\frac{\partial \mathcal{L}}{\partial W}$$

3_1 3_2

$$\frac{\partial z_1^3}{\partial W} = \begin{bmatrix} z_1^2 & z_2^2 & z_3^2 & z_4^2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial z_2^3}{\partial W} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ z_1^2 & z_2^2 & z_3^2 & z_4^2 \end{bmatrix}$$



Backward pass

Chaîne rule

$$\frac{\partial \mathscr{L}}{\partial W} = \left(\frac{\partial \mathscr{L}}{\partial \hat{\mathbf{y}}}\right)^T \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}^3} \frac{\partial \mathbf{z}^3}{\partial W}$$

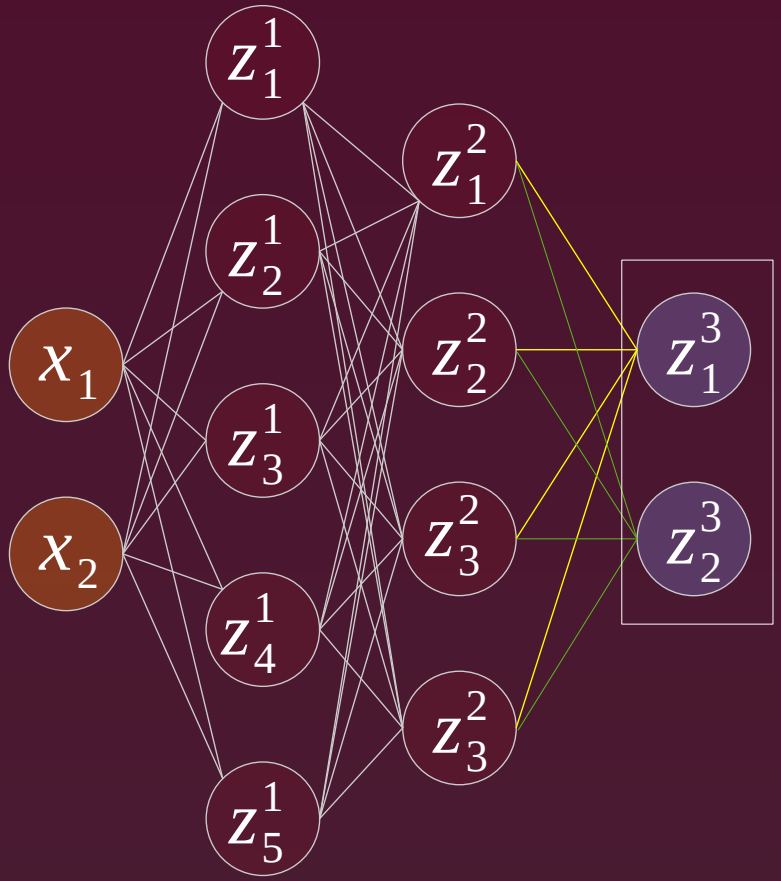
$$\left(\frac{\partial \mathscr{L}}{\partial \hat{\mathbf{y}}}\right)^T = \begin{bmatrix} \frac{-y_1}{\hat{y}_1} & \frac{-y_2}{\hat{y}_2} \end{bmatrix} \qquad \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}^3} = \begin{bmatrix} \hat{y}_1(1-\hat{y}_1) & -\hat{y}_1\hat{y}_2 \\ -\hat{y}_2\hat{y}_1 & \hat{y}_2(1-\hat{y}_2) \end{bmatrix}$$

$$\left(\frac{\partial \mathscr{L}}{\partial \hat{\mathbf{y}}}\right)^T \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}^3} = \begin{bmatrix} \frac{-y_1}{\hat{y}_1}\hat{y}_1(1-\hat{y}_1) + \frac{y_2}{\hat{y}_2}(\hat{y}_2\hat{y}_1) & \frac{y_1}{\hat{y}_1}(\hat{y}_1\hat{y}_2) - \frac{y_2}{\hat{y}_2}\hat{y}_2(1-\hat{y}_2) \end{bmatrix}$$

$$\left(\frac{\partial \mathscr{L}}{\partial \hat{\mathbf{y}}}\right)^T \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}^3} = \begin{bmatrix} -y_1 + y_1\hat{y}_1 + y_2\hat{y}_1 & y_1\hat{y}_2 - y_2 + y_2\hat{y}_2 \end{bmatrix}$$

$$\left(\frac{\partial \mathscr{L}}{\partial \hat{\mathbf{y}}}\right)^T \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}^3} = \begin{bmatrix} -y_1 + (y_1+y_2)\hat{y}_1 & -y_2 + \hat{y}_2(y_1+y_2) \end{bmatrix}$$

$$\left(\frac{\partial \mathscr{L}}{\partial \hat{\mathbf{y}}}\right)^T \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}^3} = \begin{bmatrix} \hat{y}_1 - y_1 & \hat{y}_2 - y_2 \end{bmatrix}; \text{ car } y_1 + y_2 = 1$$



Backward pass

Chaîne rule

$$\frac{\partial \mathscr{L}}{\partial W} = \left(\frac{\partial \mathscr{L}}{\partial \hat{y}}\right)^T \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\frac{\partial z_1^3}{\partial W} = \begin{bmatrix} z_1^2 & z_2^2 & z_3^2 & z_4^2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \frac{\partial z_2^3}{\partial W} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ z_1^2 & z_2^2 & z_3^2 & z_4^2 \end{bmatrix}$$

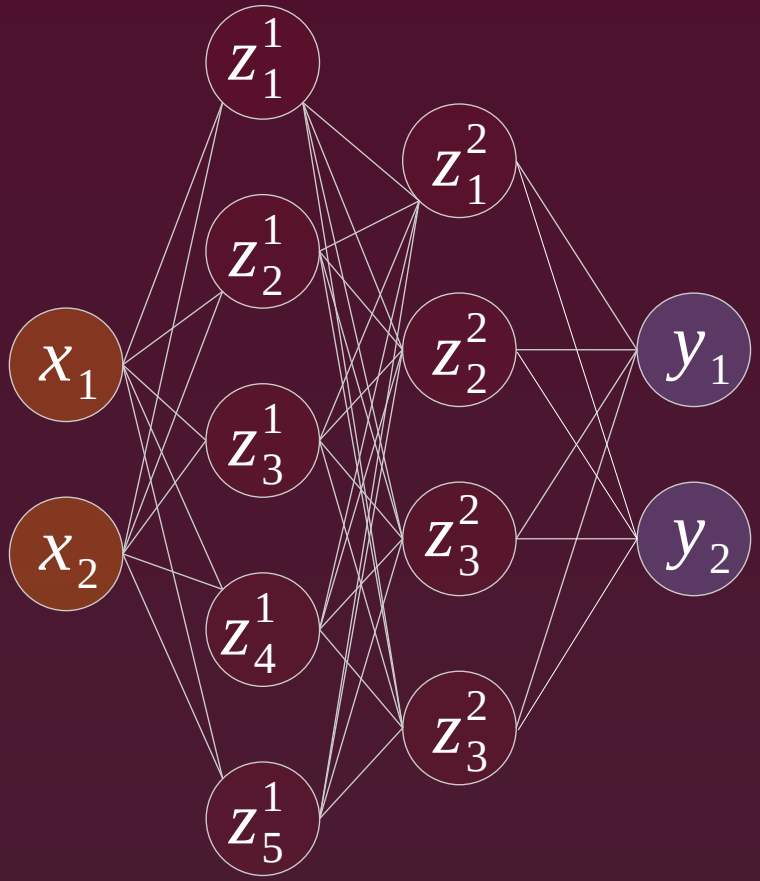
$$\left(\frac{\partial \mathscr{L}}{\partial \hat{y}}\right)^T \frac{\partial \hat{y}}{\partial z^3} = \begin{bmatrix} \hat{y}_1 - y_1 & \hat{y}_2 - y_2 \end{bmatrix}; \text{ car } y_1 + y_2 = 1$$

$$\frac{\partial \mathscr{L}}{\partial W} = (\hat{y}_1 - y_1) \begin{bmatrix} z_1^2 & z_2^2 & z_3^2 & z_4^2 \\ 0 & 0 & 0 & 0 \end{bmatrix} + (\hat{y}_2 - y_2) \begin{bmatrix} 0 & 0 & 0 & 0 \\ z_1^2 & z_2^2 & z_3^2 & z_4^2 \end{bmatrix}$$

$$\frac{\partial \mathscr{L}}{\partial W} = \begin{bmatrix} (\hat{y}_1 - y_1) z_1^2 & (\hat{y}_1 - y_1) z_2^2 & (\hat{y}_1 - y_1) z_3^2 & (\hat{y}_1 - y_1) z_4^2 \\ (\hat{y}_2 - y_2) z_1^2 & (\hat{y}_2 - y_2) z_2^2 & (\hat{y}_2 - y_2) z_3^2 & (\hat{y}_2 - y_2) z_4^2 \end{bmatrix}$$

$$\frac{\partial \mathscr{L}}{\partial W} = (\hat{y} - y)(z^2)^T; \; z^2 = [z_1^2, z_2^2, z_3^2, z_4^2]^T$$

Implémentation en python

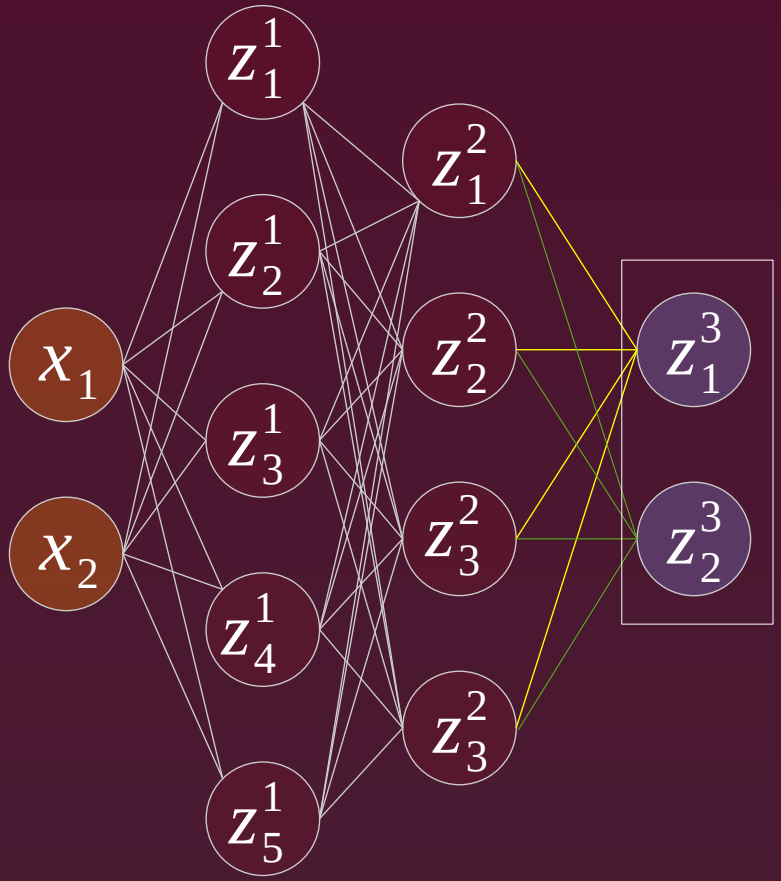


Calcul de gradients à l'aide de TensorFlow et PyTorch , deux bibliothèques Python de deep learning.

TensorFlow

PyTorch

Forward pass



Backward pass

Chaîne rule

$$\frac{\partial \mathcal{L}}{\partial W} = \left(\frac{\partial \mathcal{L}}{\partial \hat{y}}\right)^T \frac{\partial \hat{y}}{\partial z^3} \frac{\partial z^3}{\partial W}$$

$$\frac{\partial \mathcal{L}}{\partial W} = (\hat{y} - y)(z^2)^T ;\ z^2 = [z_1^2, z_2^2, z_3^2, z_4^2]^T$$

$$\frac{\partial \mathcal{L}}{\partial W} = (\hat{y} - y)(z^2)^T ;\ z^2 = [z_1^2, z_2^2, z_3^2, z_4^2]^T$$

Pas d'apprentissage

$$W = W - \ə \frac{\partial \mathcal{L}}{\partial W}$$

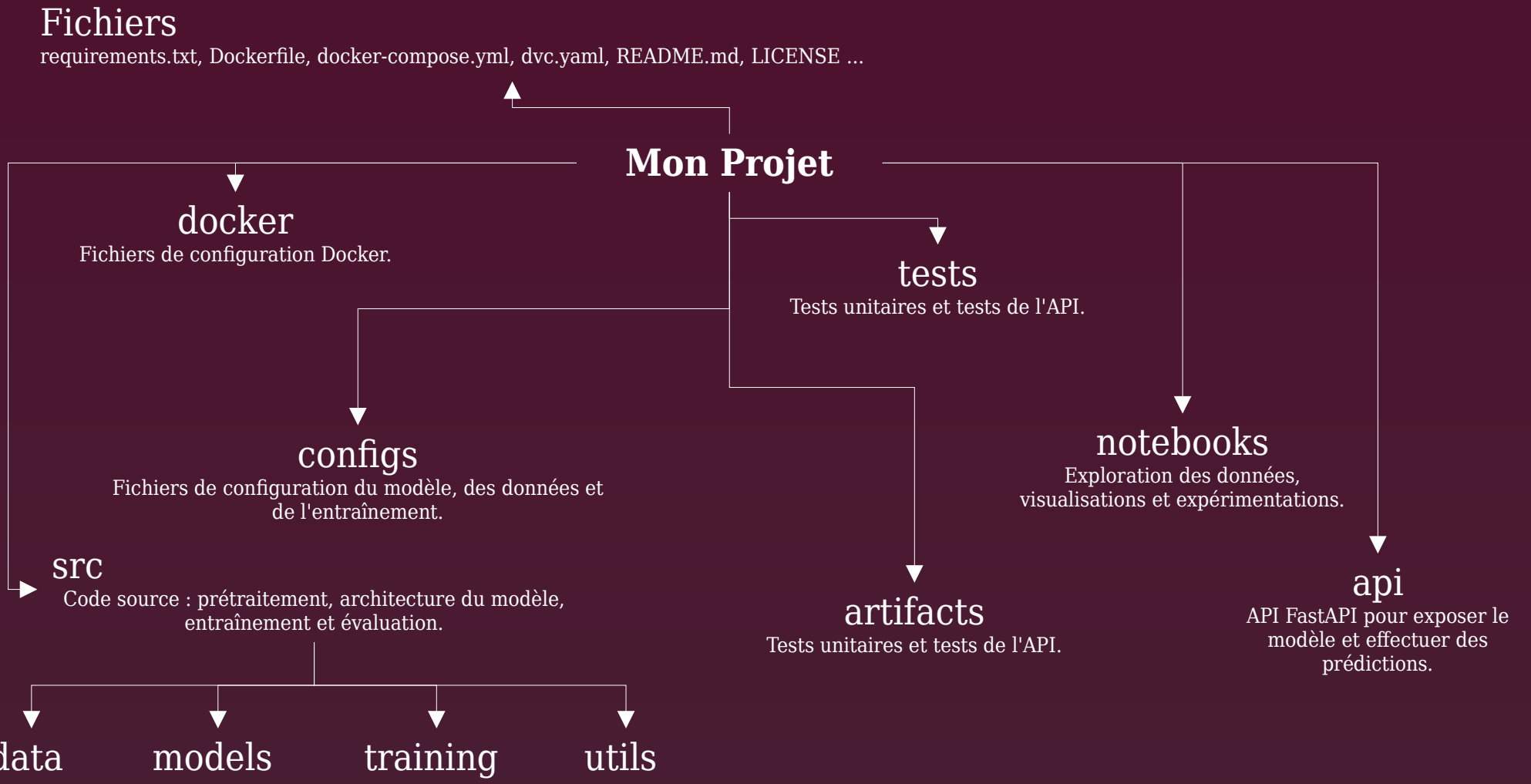
Nouvelle valeur de W

Ancienne valeur de W →

Cette dérivée partielle nous indique comment modifier la valeur de w afin de minimiser l'erreur.

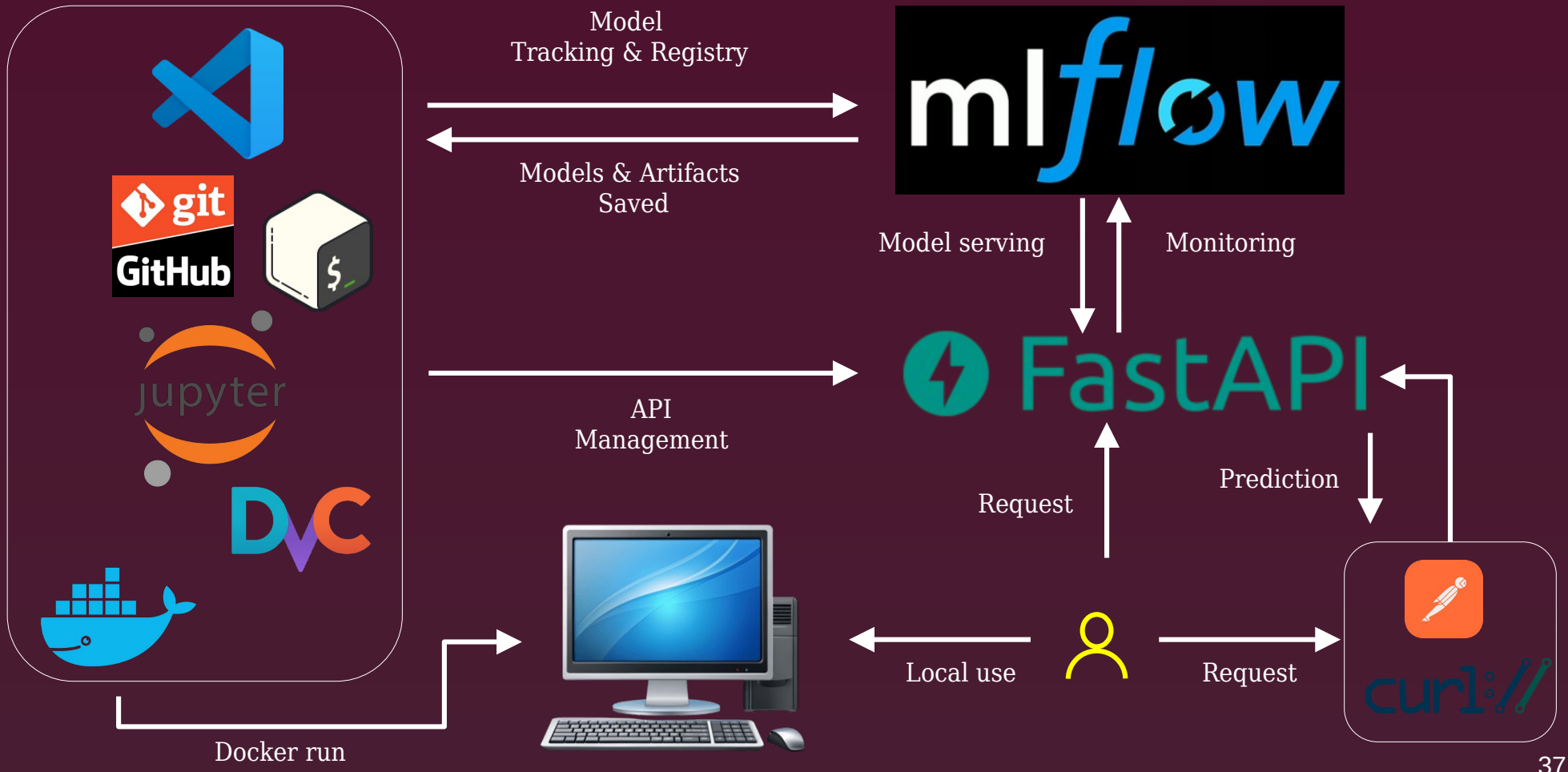
$$W = \begin{bmatrix} w_1^1 & w_2^1 & w_3^1 & w_4^1 \\ w_1^2 & w_2^2 & w_3^2 & w_2^2 \end{bmatrix}$$

Structure du projet



Déploiement

Outils de développement et de déploiement



Merci !